

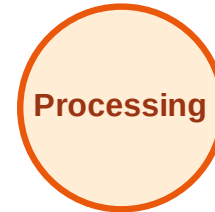
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



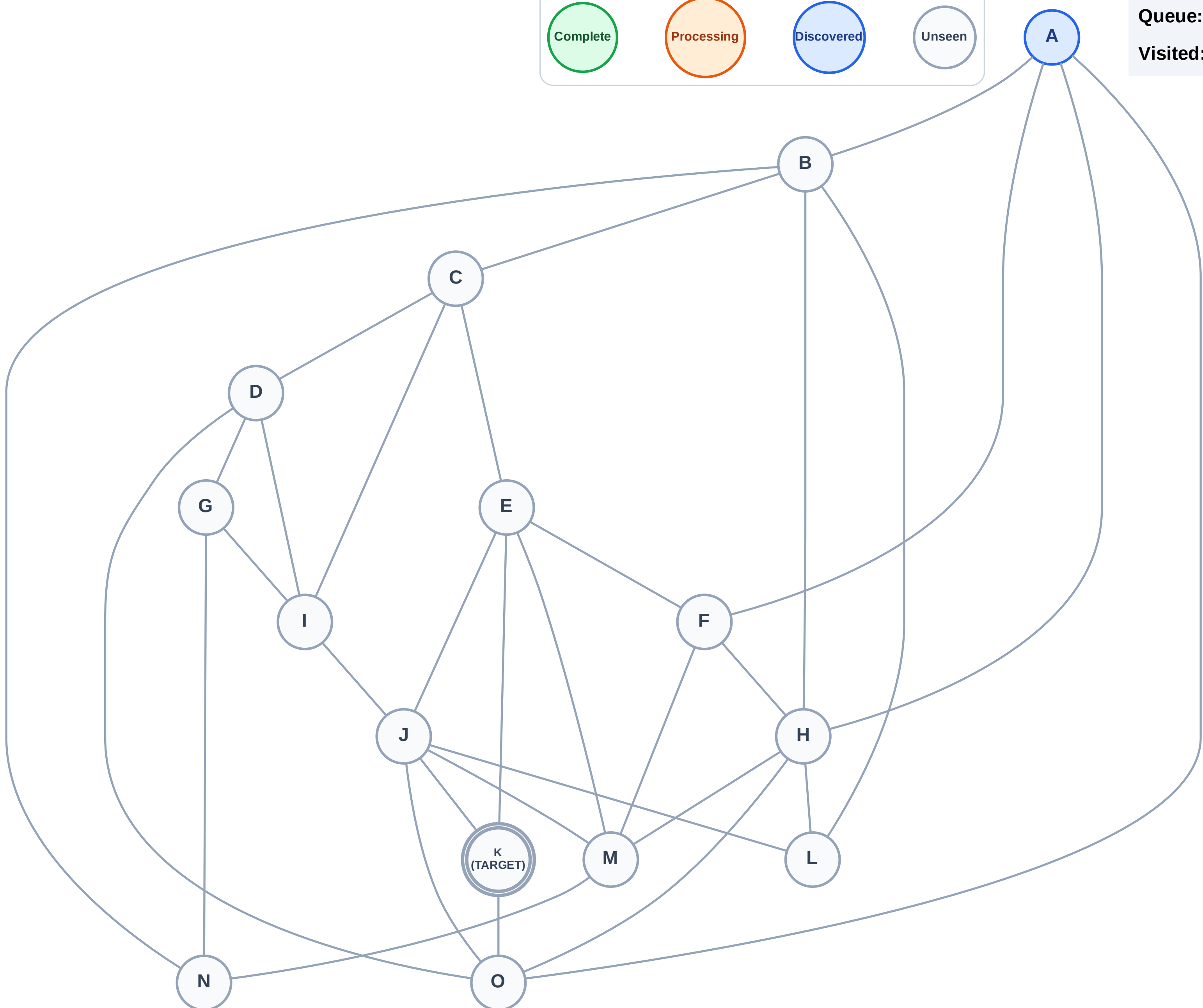
Discovered



Unseen

Queue: A

Visited: (none)



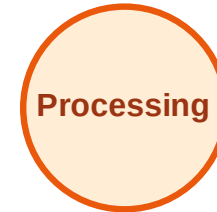
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



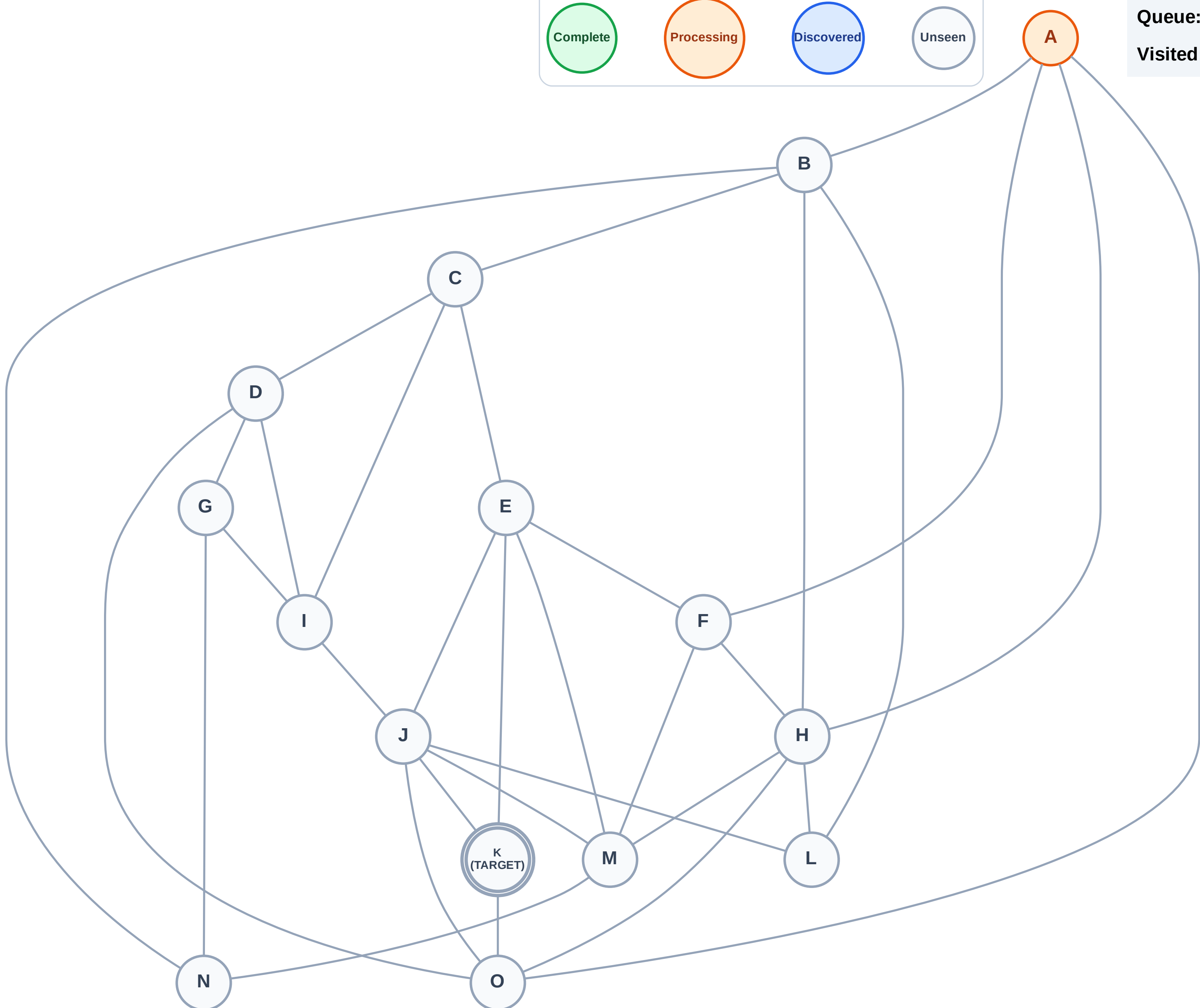
Discovered



Unseen

Queue: (empty)

Visited: (none)



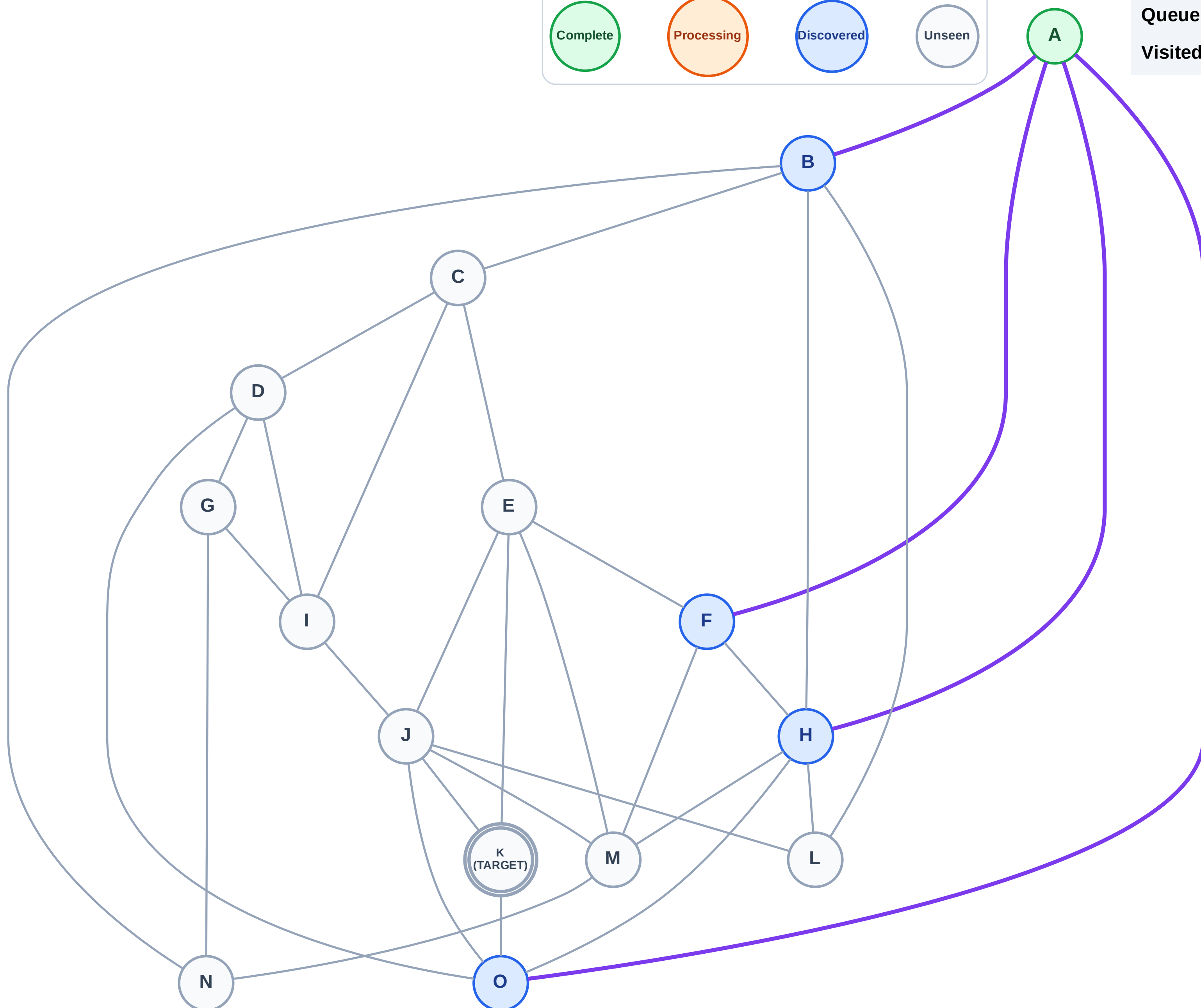
BFS Traversal

Step 3: Discover B, F, H, O from A

Legend



Queue: B → F → H → O
Visited: A



BFS Traversal

Step 4: Process node B

Legend

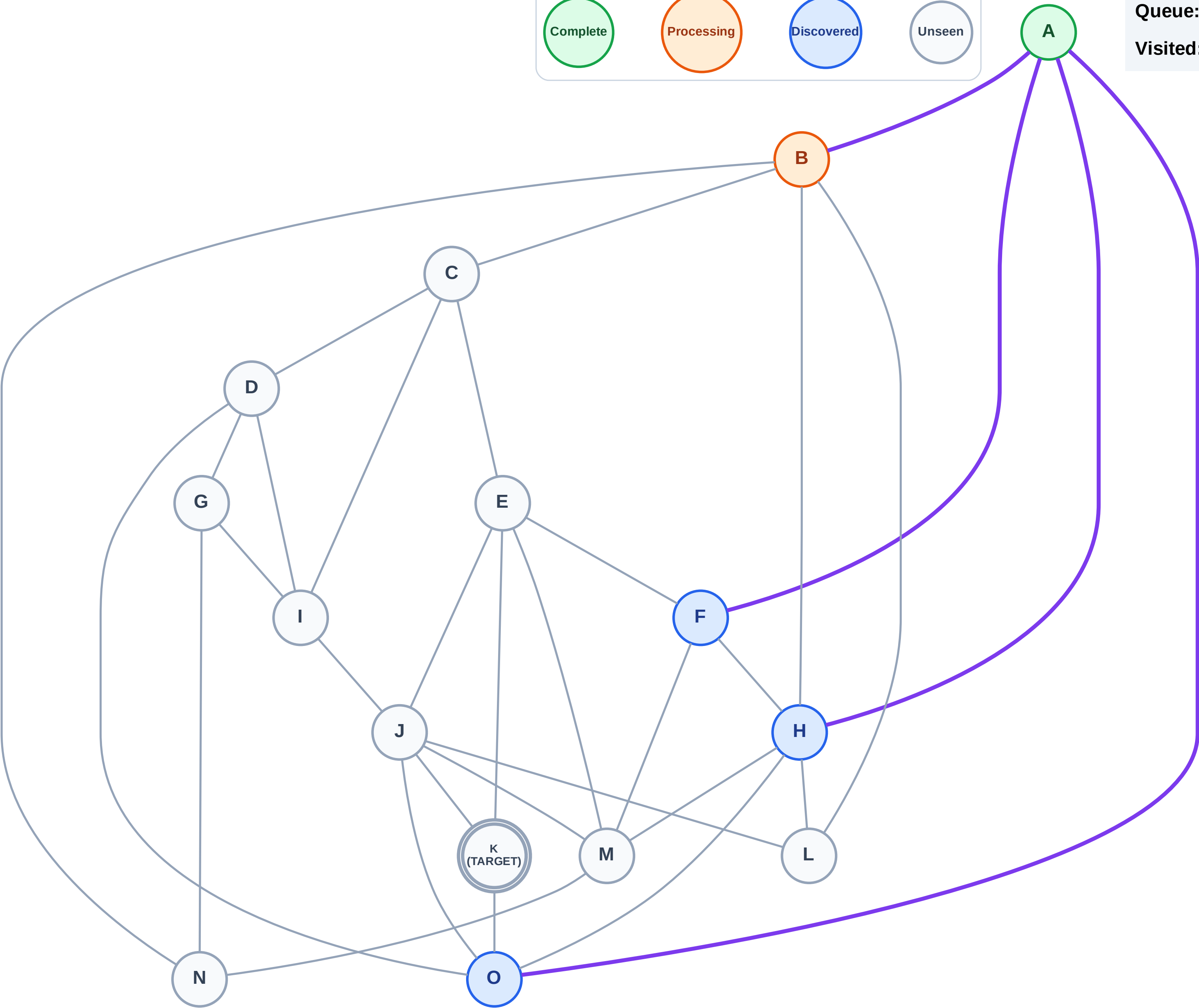
Complete

Processing

Discovered

Unseen

Queue: F → H → O
Visited: A



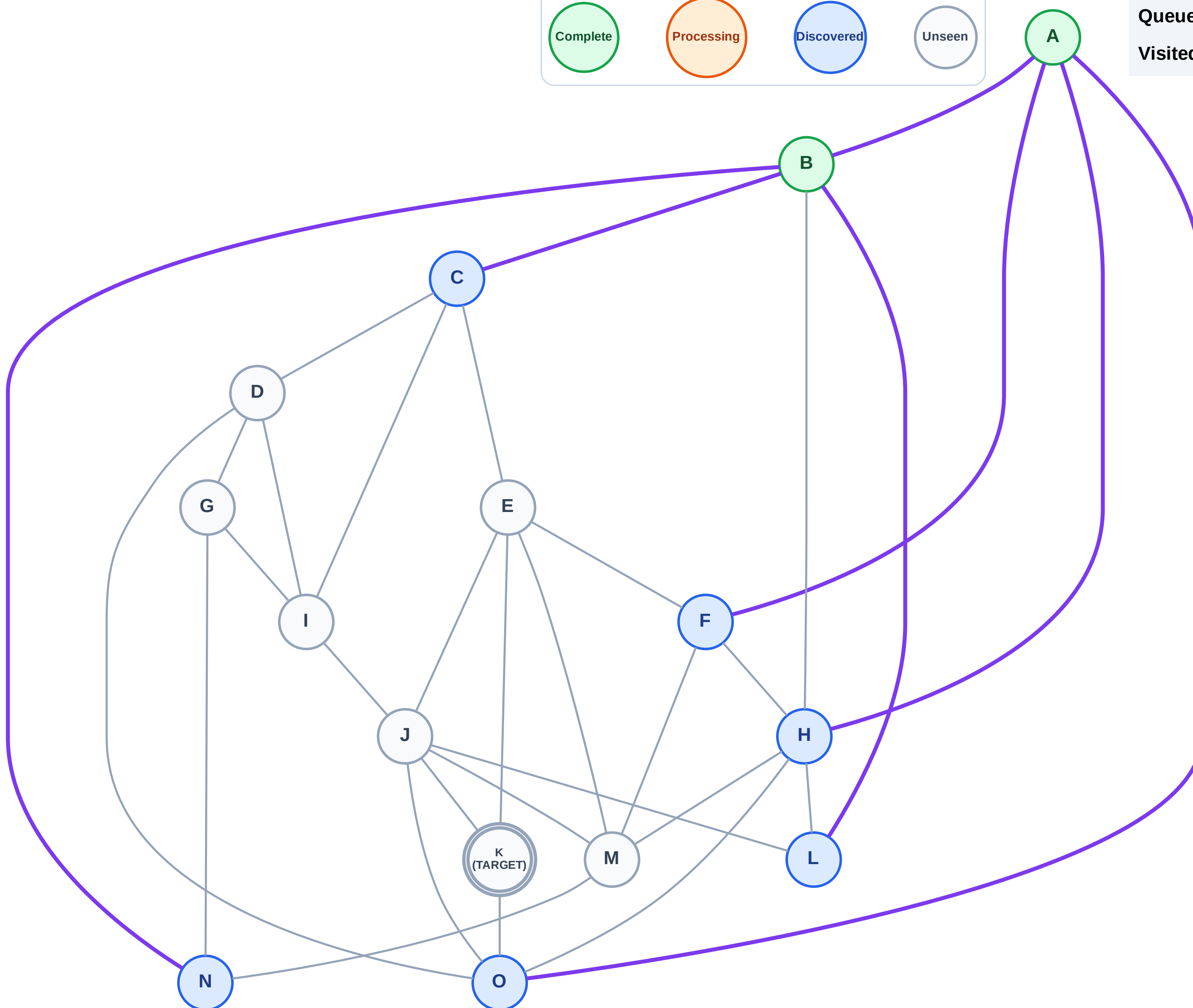
BFS Traversal

Step 5: Discover C, L, N from B

Legend



Queue: F → H → O → C → L → N
Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

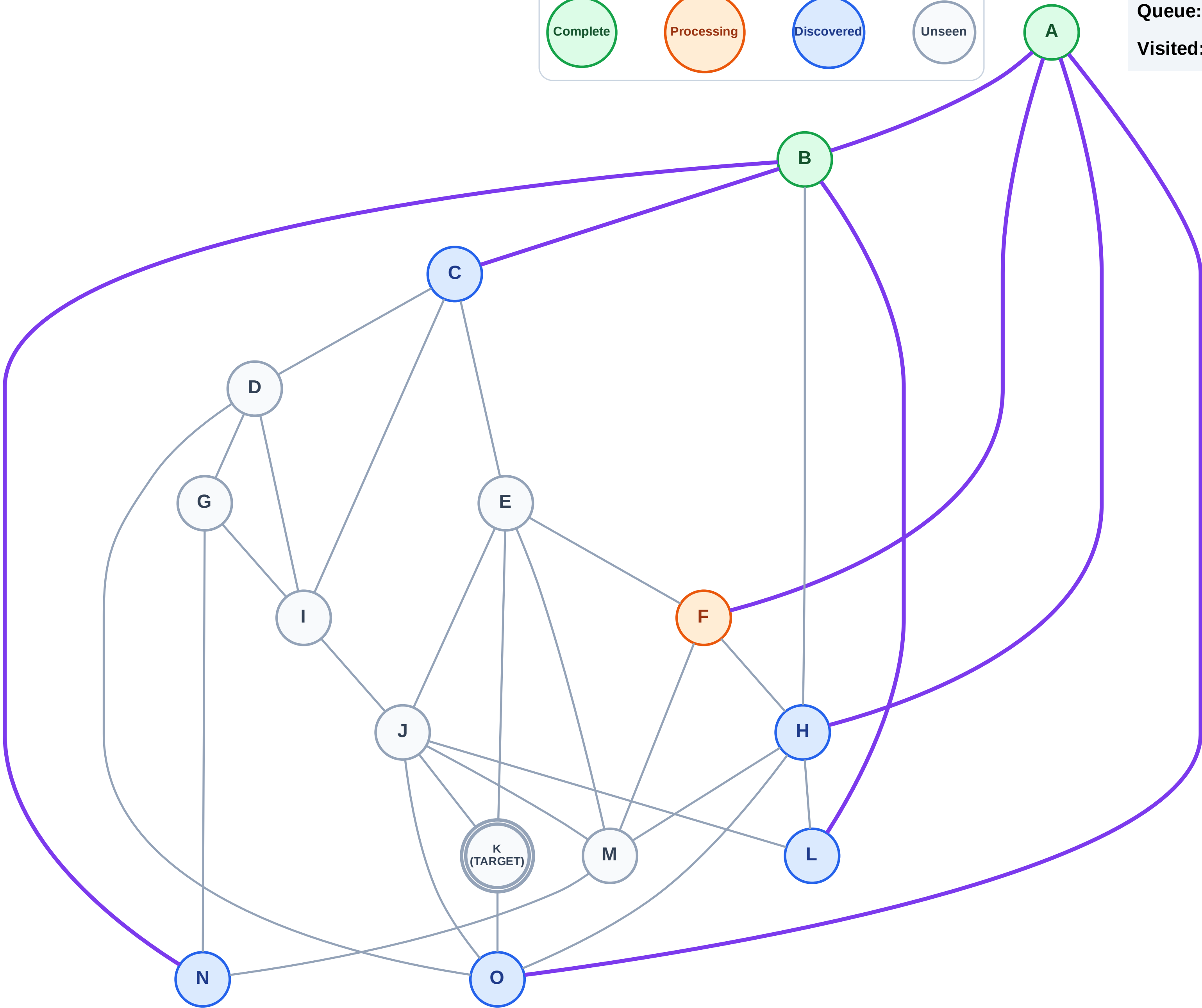
Processing

Discovered

Unseen

Queue: H → O → C → L → N

Visited: A, B



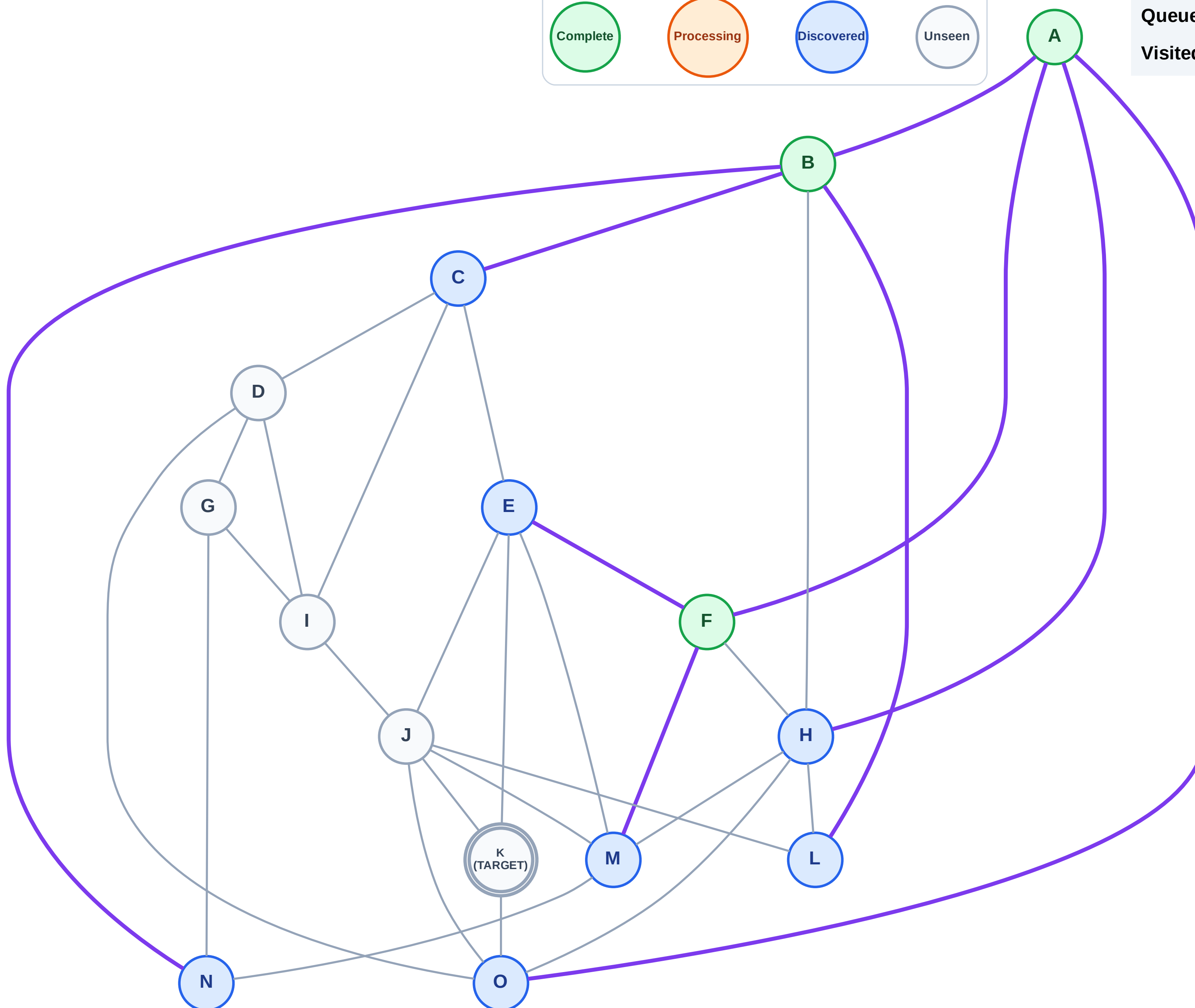
BFS Traversal

Step 7: Discover E, M from F

Legend



Queue: H → O → C → L → N → E → M
Visited: A, B, F



BFS Traversal

Step 8: Process node H

Legend

Complete

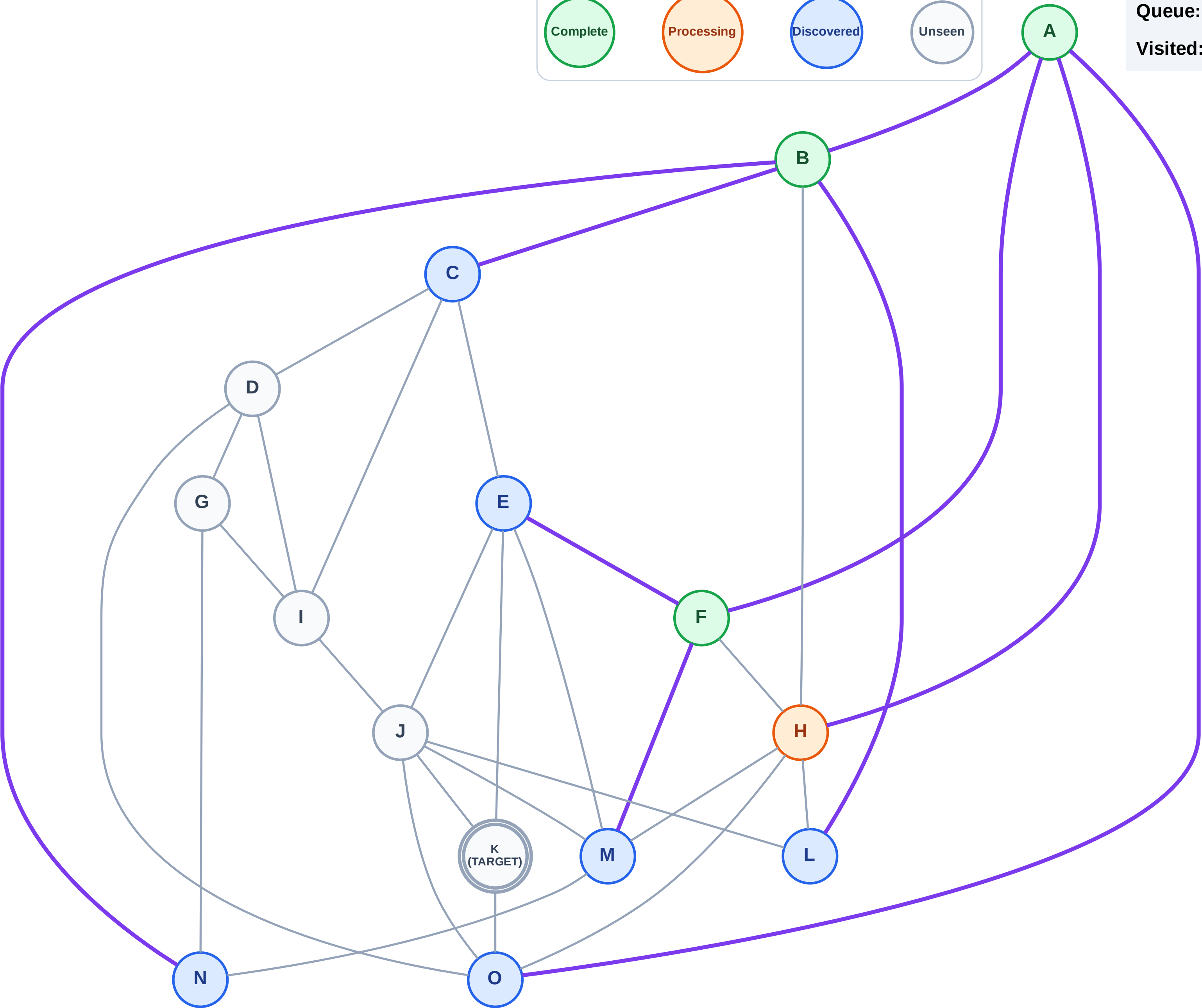
Processing

Discovered

Unseen

Queue: O → C → L → N → E → M

Visited: A, B, F



BFS Traversal

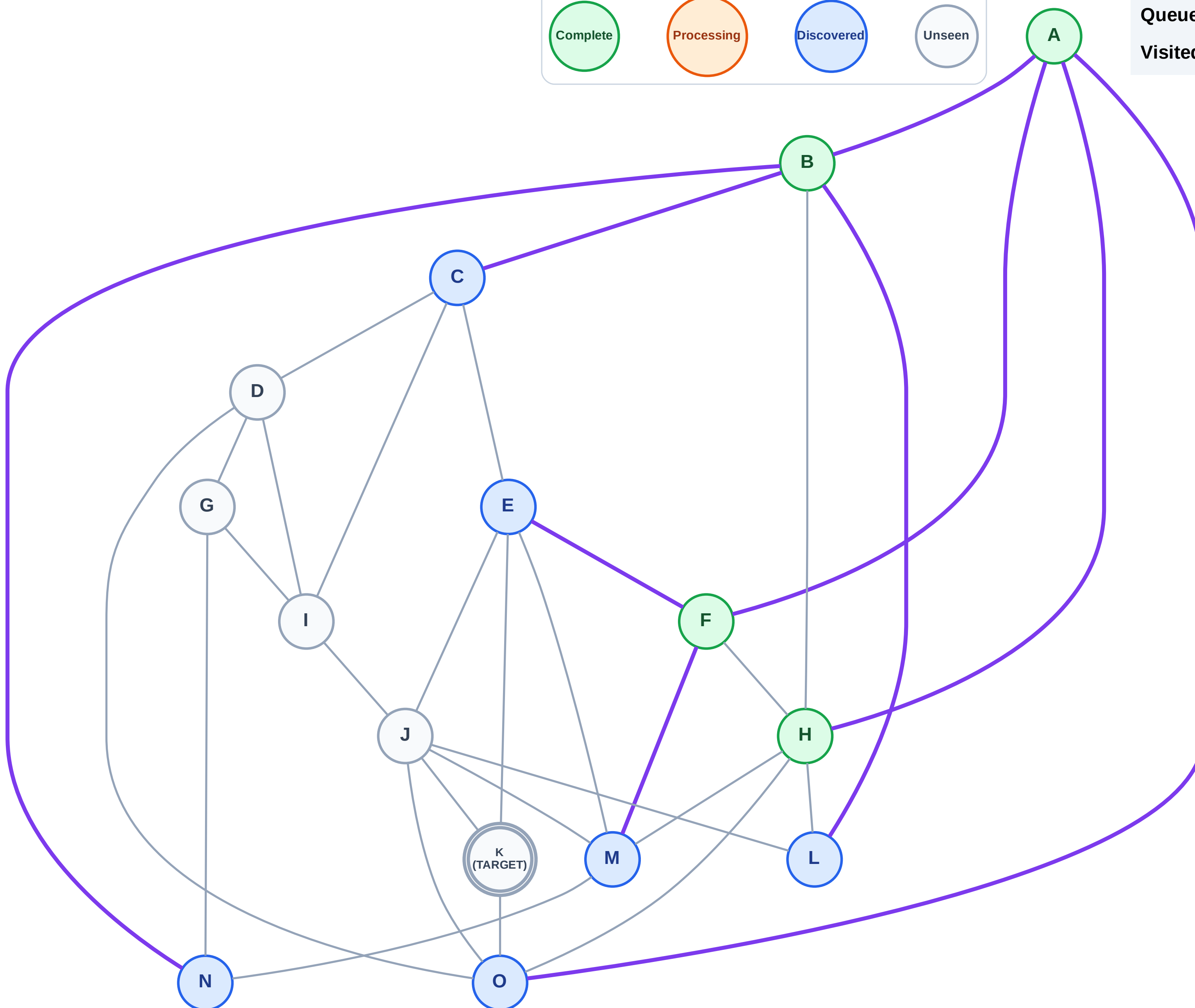
Step 9: No new neighbors from H

Legend



Queue: O → C → L → N → E → M

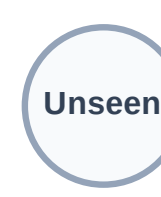
Visited: A, B, F, H



BFS Traversal

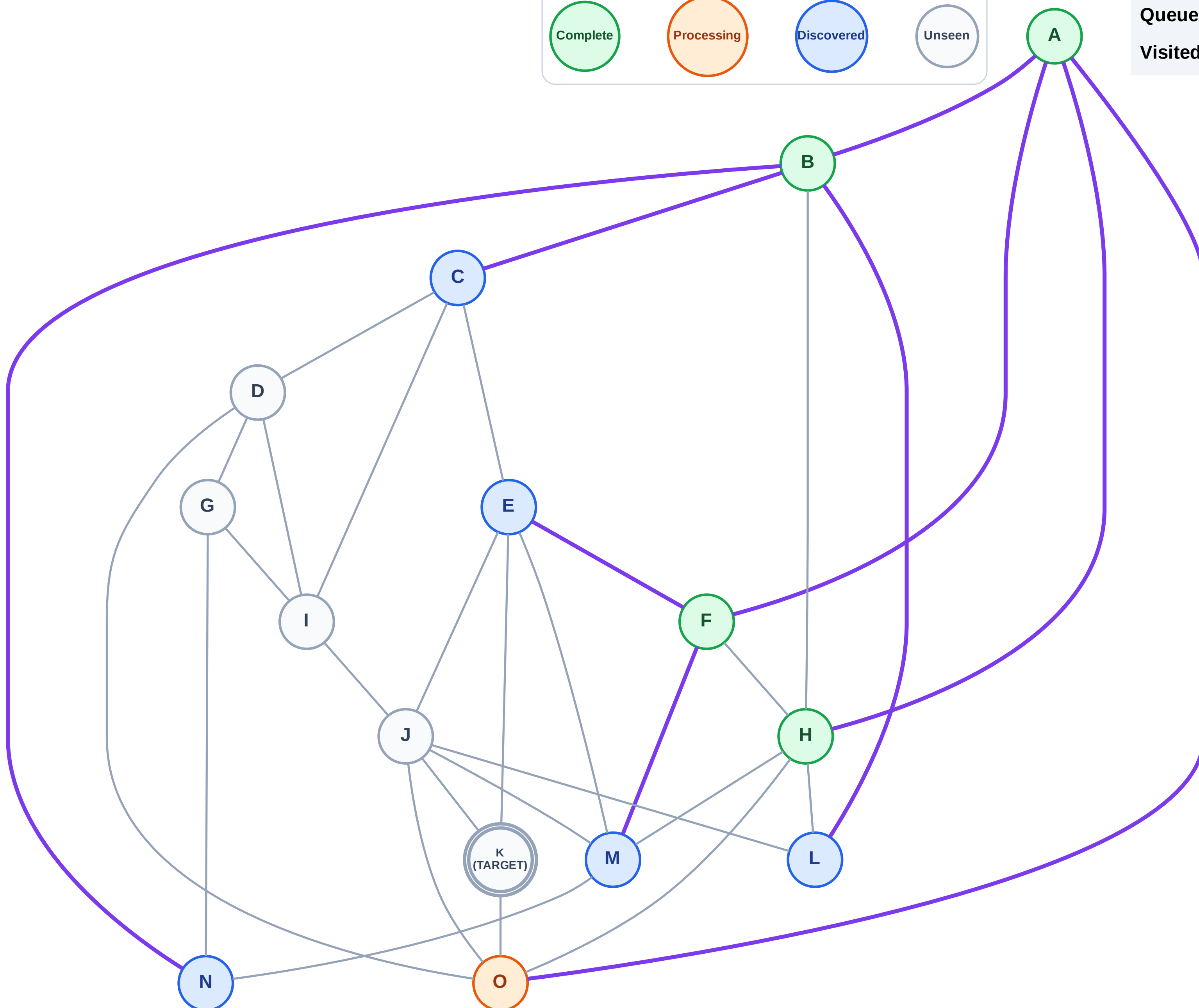
Step 10: Process node O

Legend



Queue: C → L → N → E → M

Visited: A, B, F, H



BFS Traversal

Step 11: Discover D, J, K from O

Legend

Complete

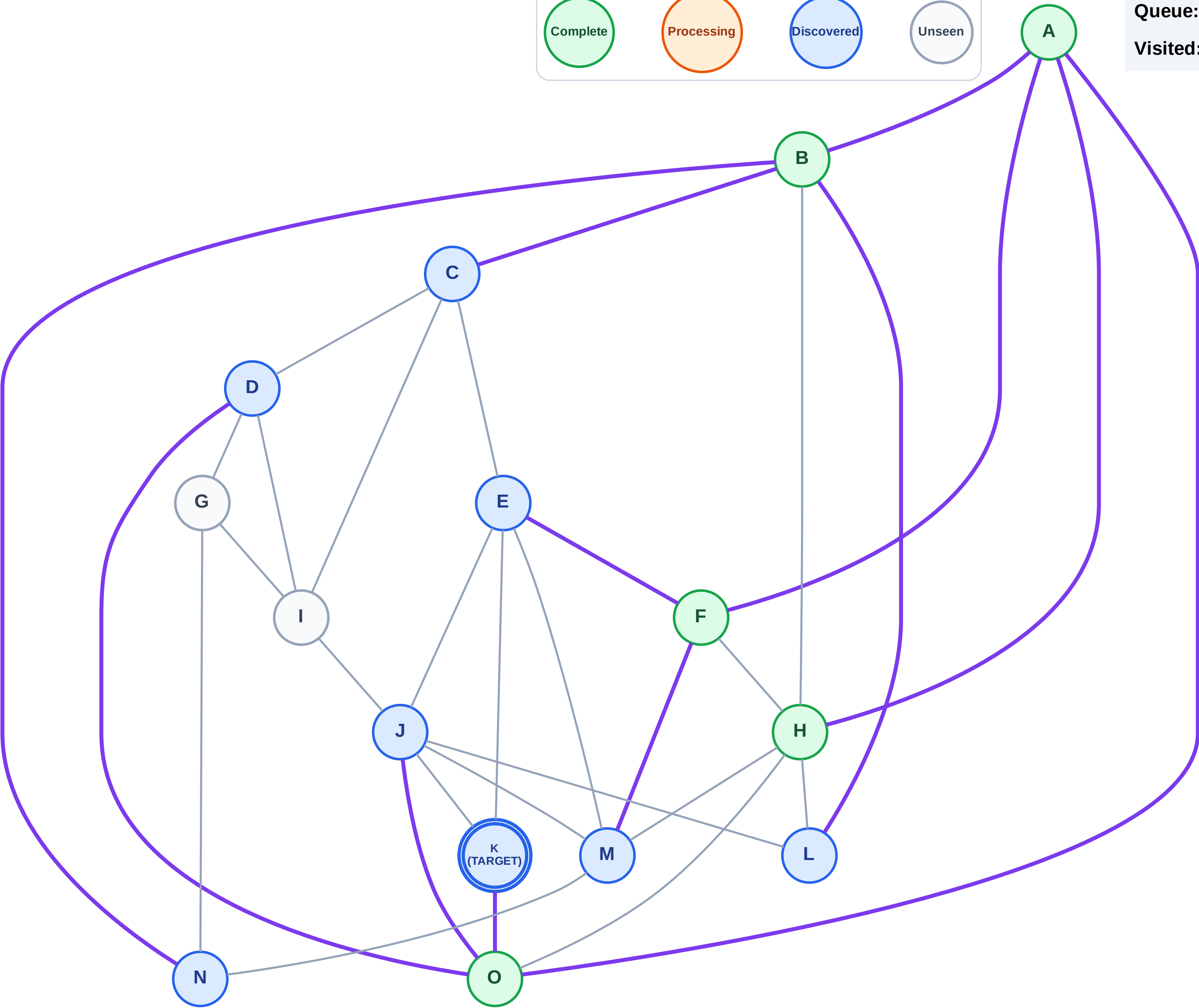
Processing

Discovered

Unseen

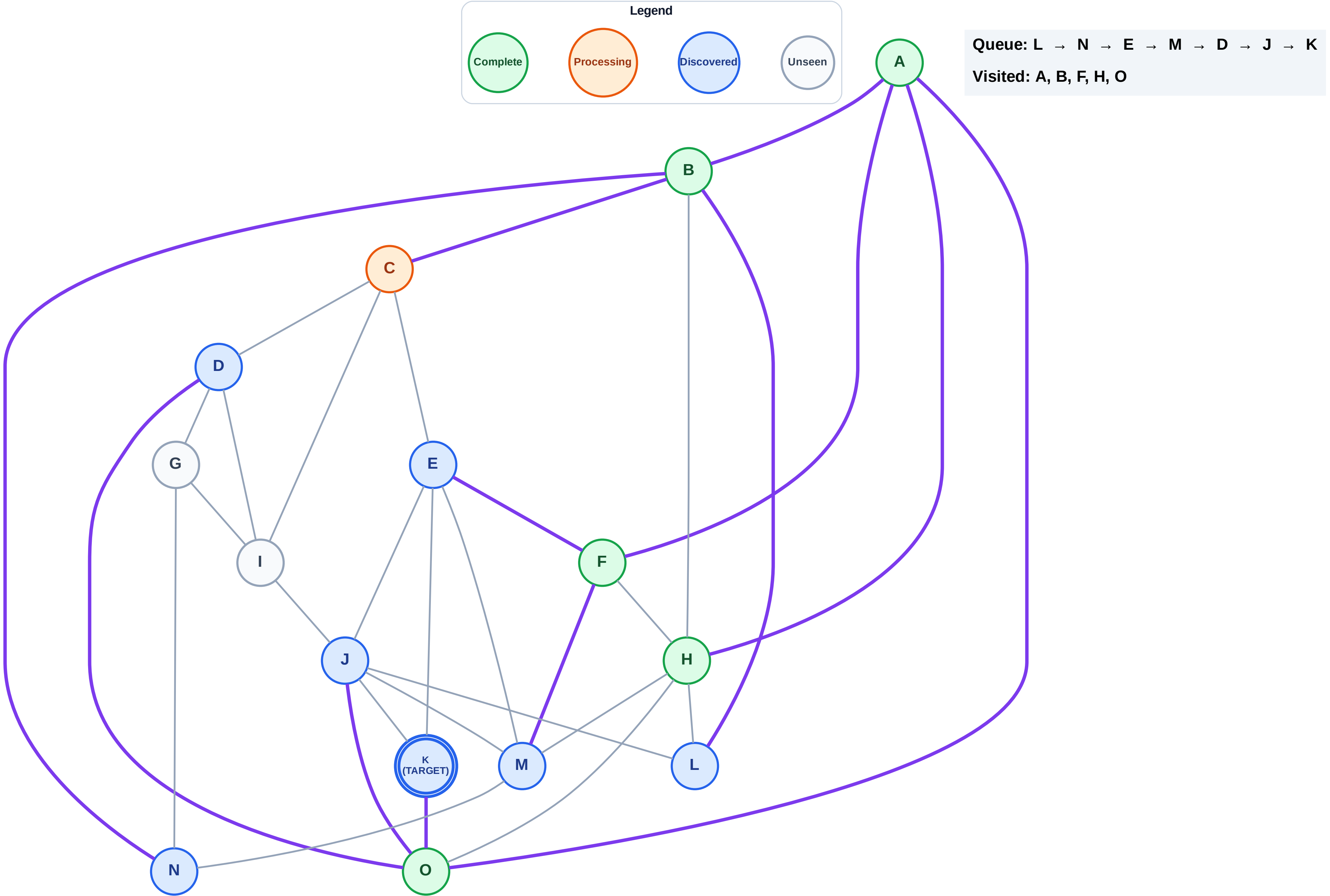
Queue: C → L → N → E → M → D → J → K

Visited: A, B, F, H, O



BFS Traversal

Step 12: Process node C



BFS Traversal

Step 13: Discover I from C

Legend



Complete



Processing



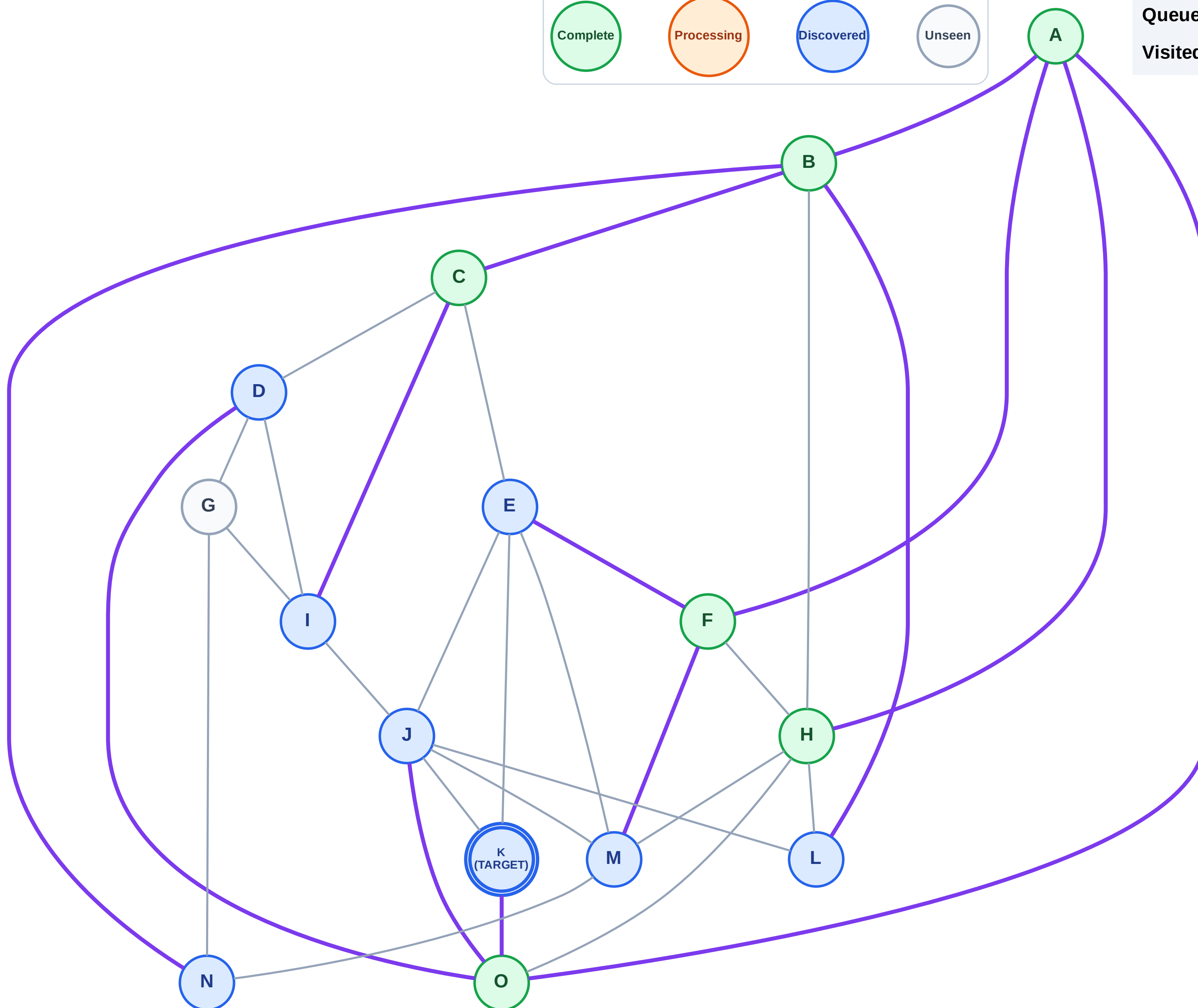
Discovered



Unseen

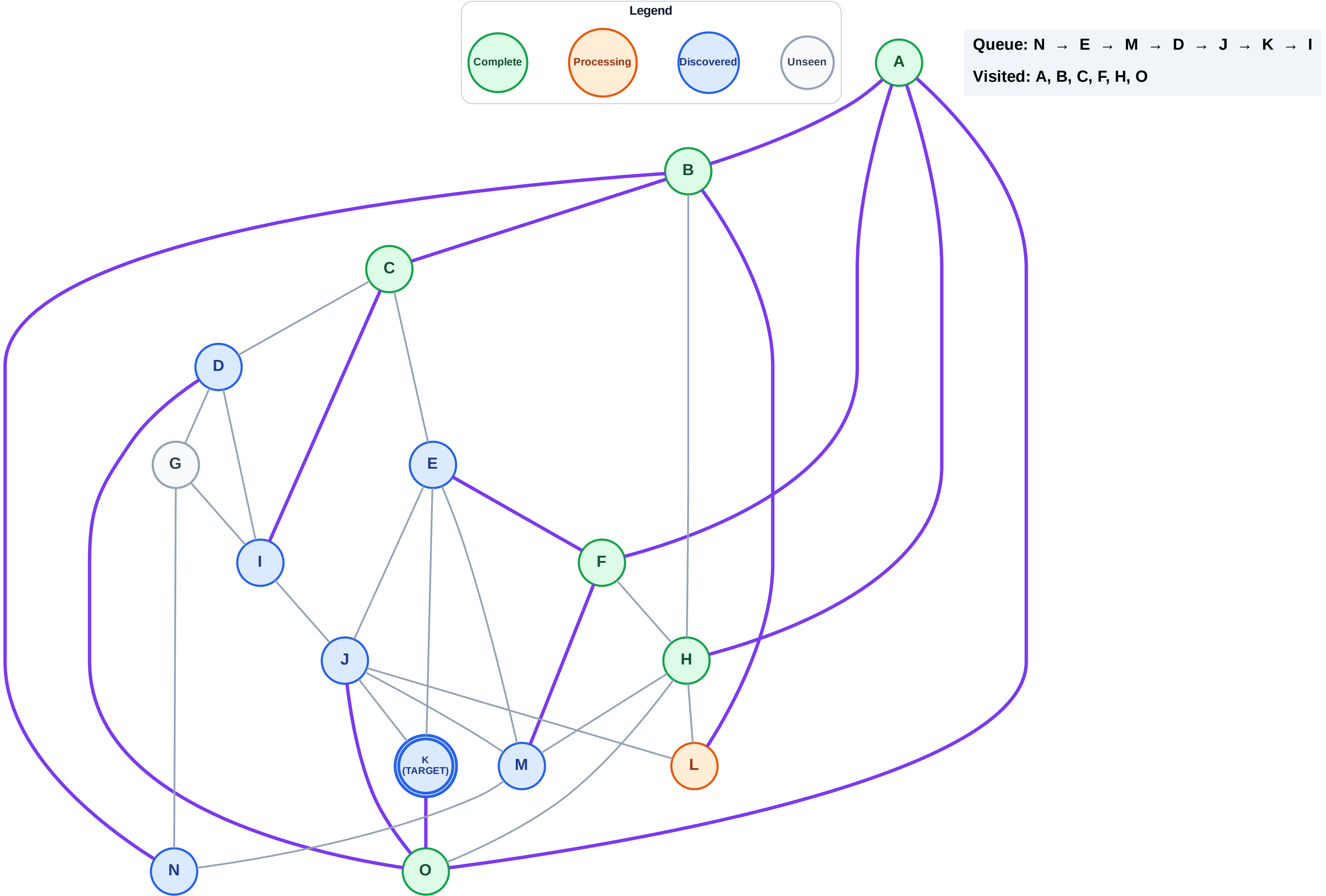
Queue: L → N → E → M → D → J → K → I

Visited: A, B, C, F, H, O



BFS Traversal

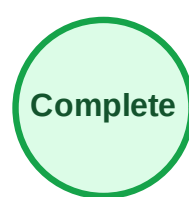
Step 14: Process node L



BFS Traversal

Step 15: No new neighbors from L

Legend



Complete



Processing

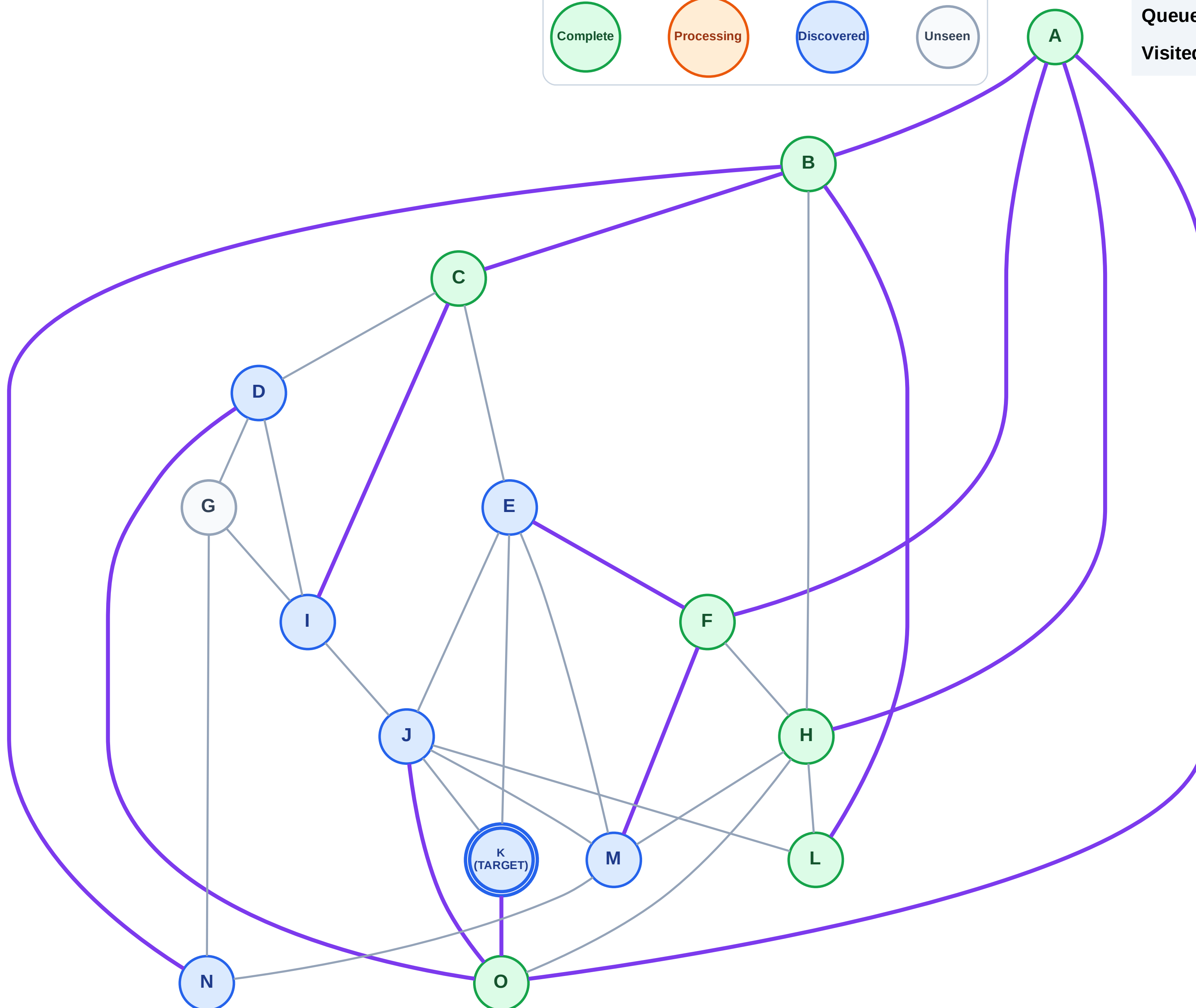


Discovered



Unseen

Queue: N → E → M → D → J → K → I
Visited: A, B, C, F, H, L, O



BFS Traversal

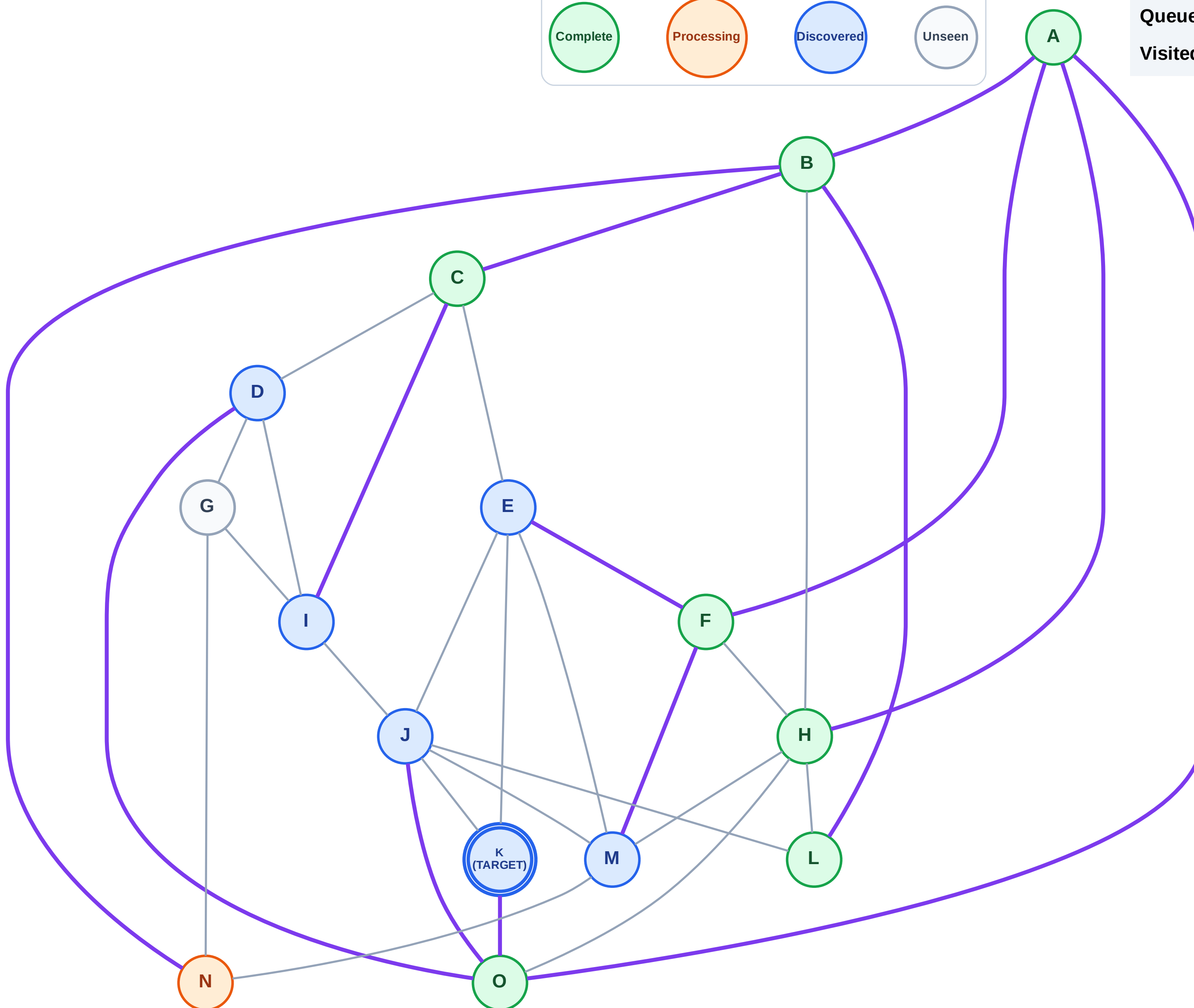
Step 16: Process node N

Legend



Queue: E → M → D → J → K → I

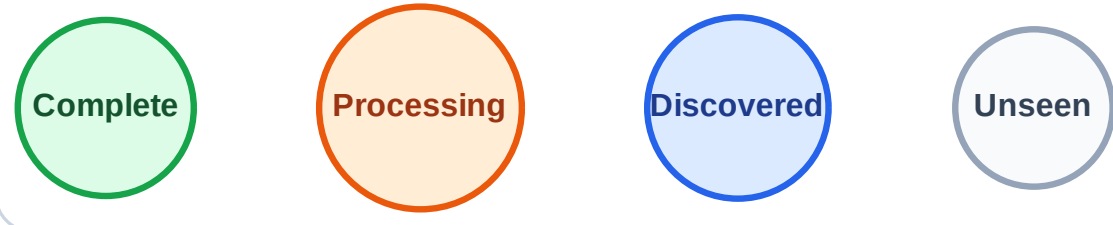
Visited: A, B, C, F, H, L, O



BFS Traversal

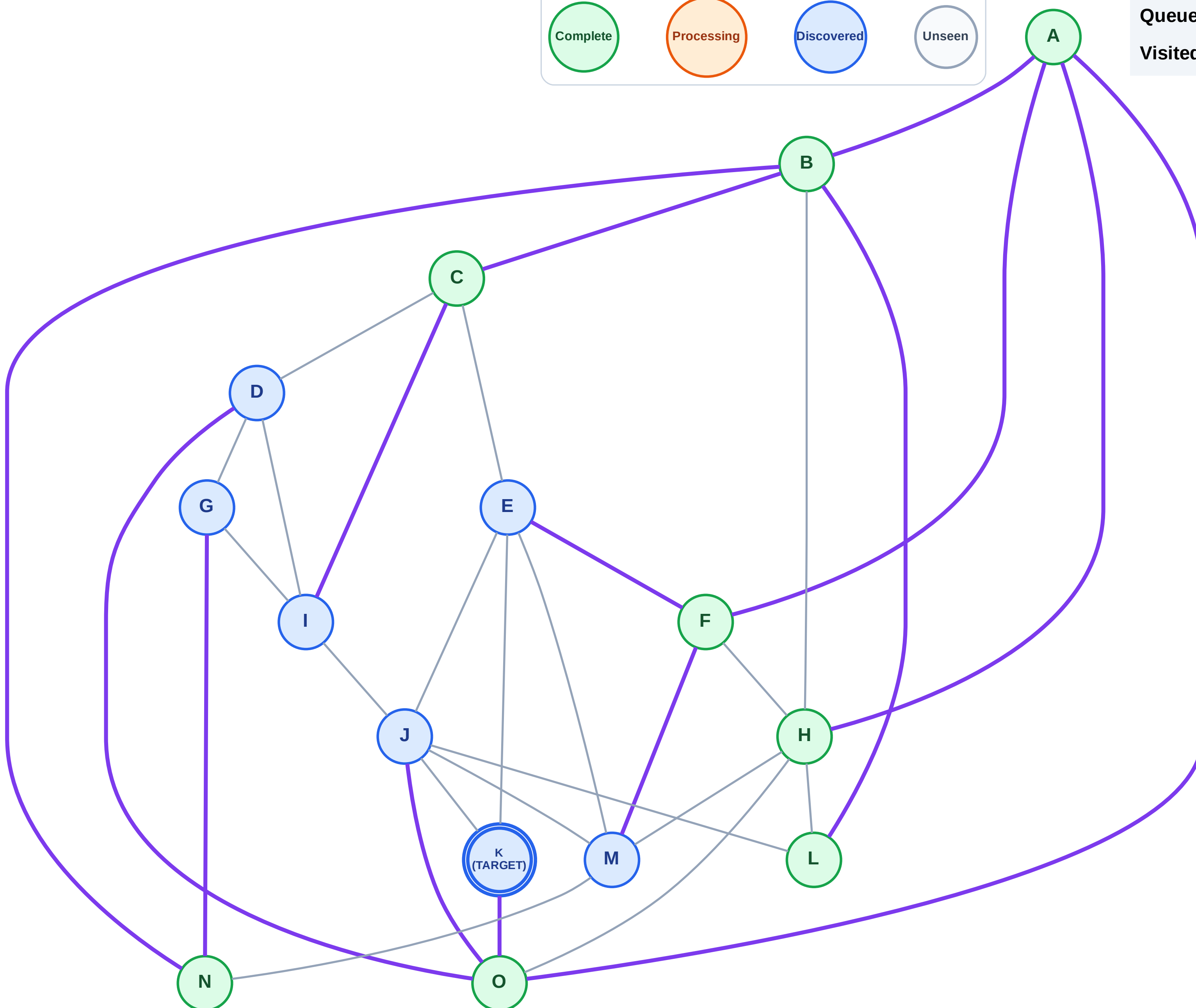
Step 17: Discover G from N

Legend



Queue: E → M → D → J → K → I → G

Visited: A, B, C, F, H, L, N, O



BFS Traversal

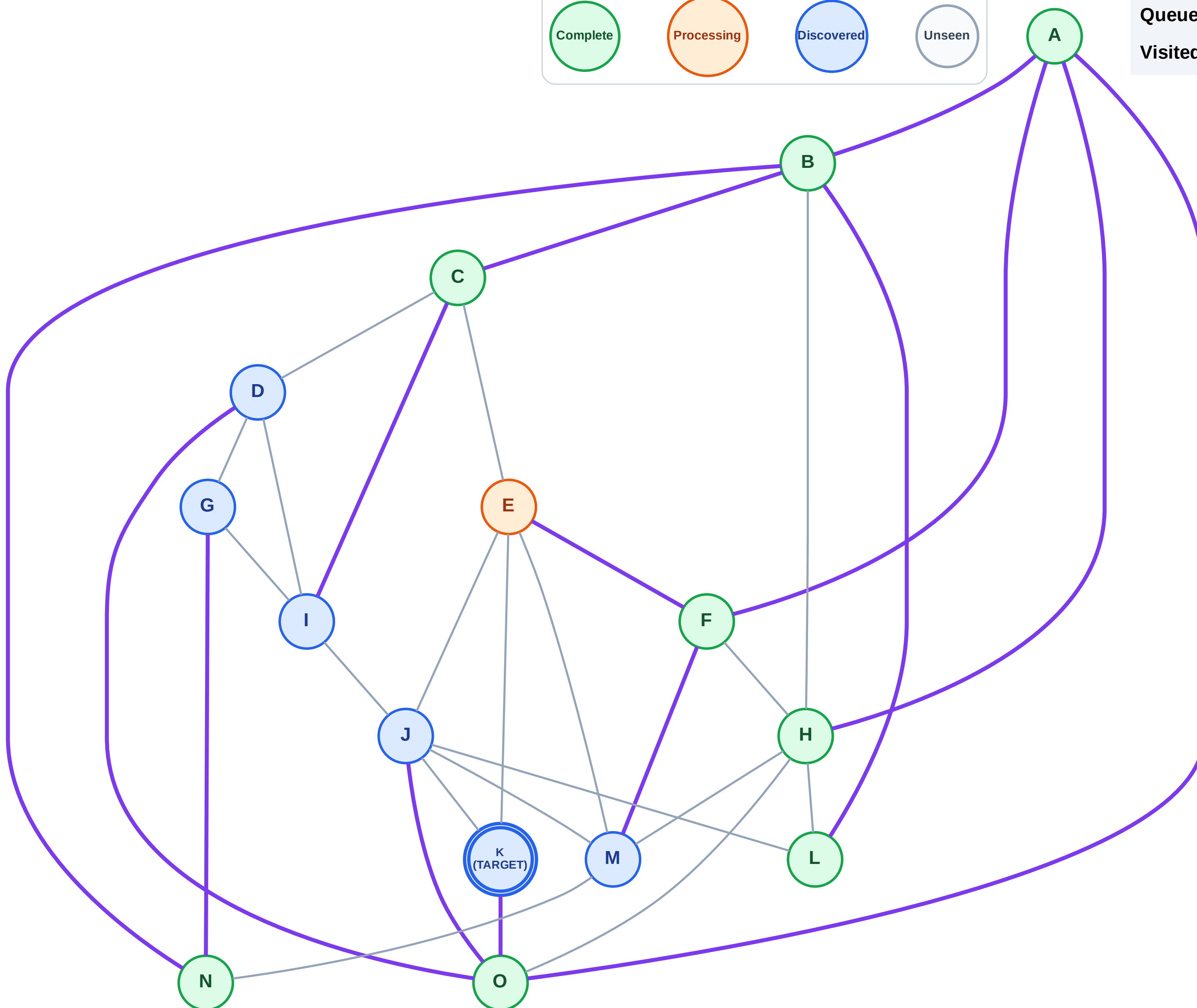
Step 18: Process node E

Legend



Queue: M → D → J → K → I → G

Visited: A, B, C, F, H, L, N, O



BFS Traversal

Step 19: No new neighbors from E

Legend

Complete

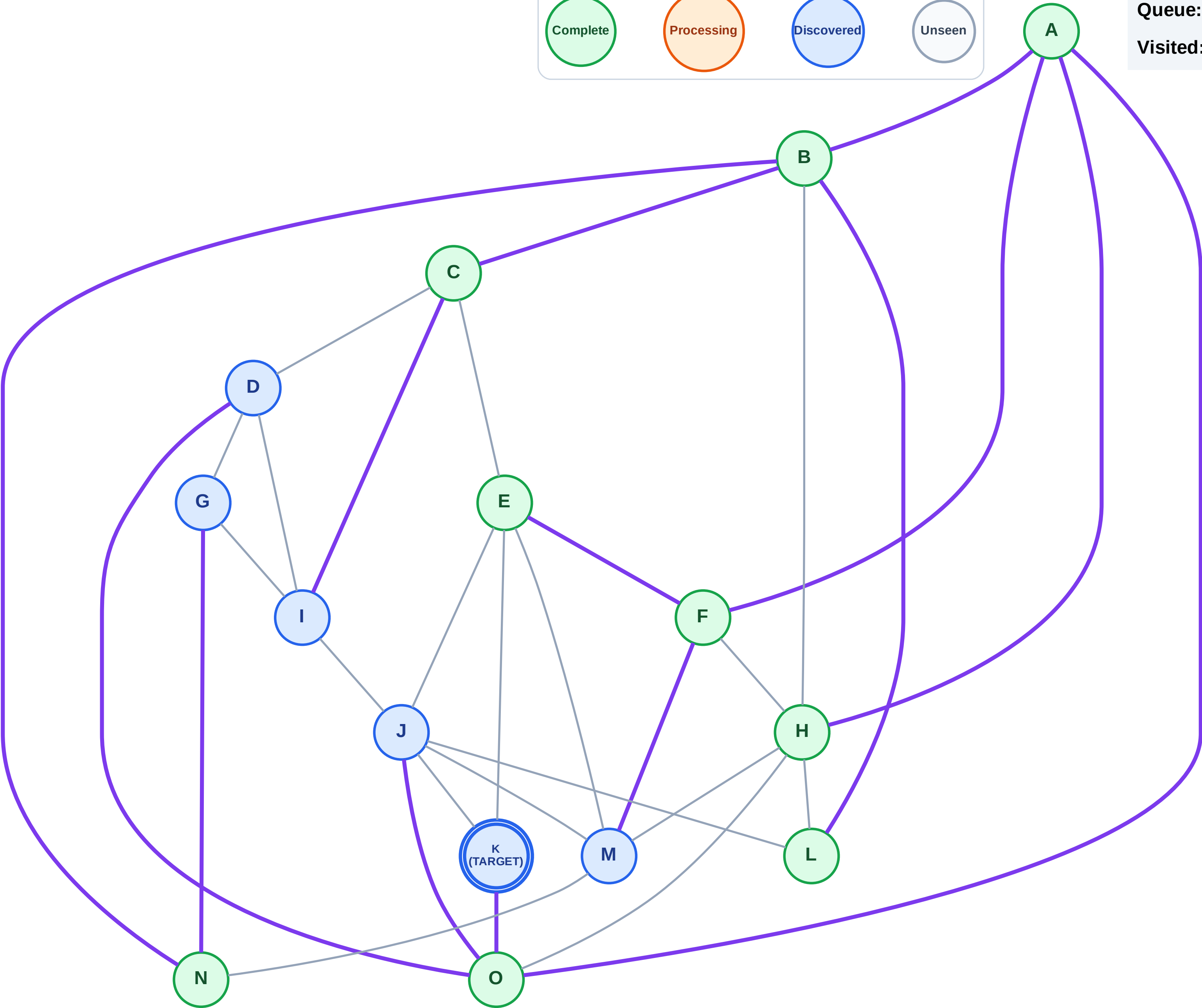
Processing

Discovered

Unseen

Queue: M → D → J → K → I → G

Visited: A, B, C, E, F, H, L, N, O



BFS Traversal

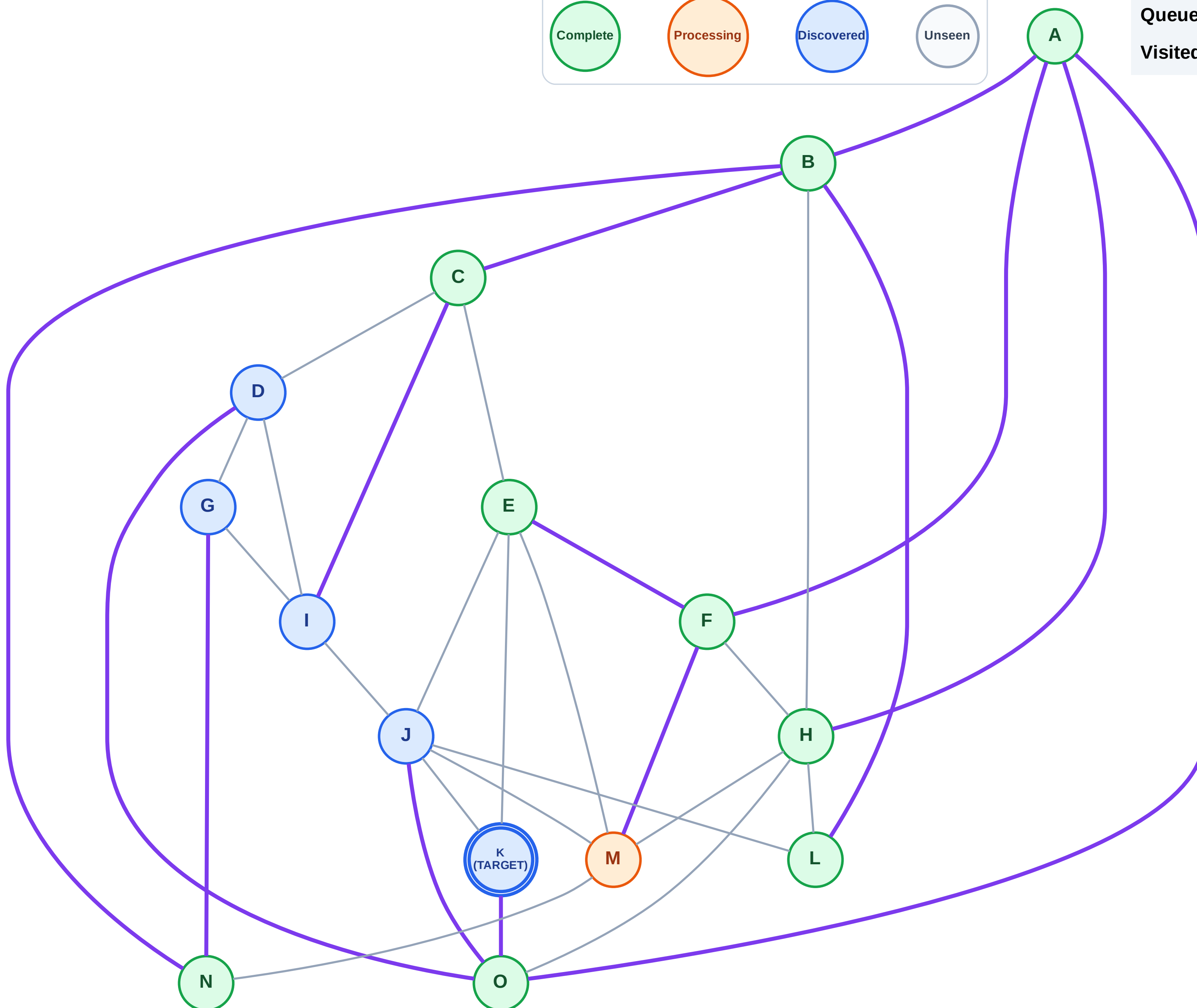
Step 20: Process node M

Legend



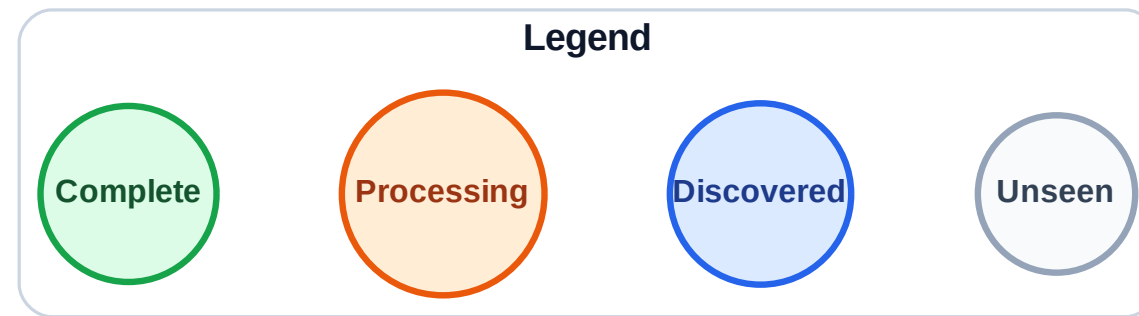
Queue: D → J → K → I → G

Visited: A, B, C, E, F, H, L, N, O

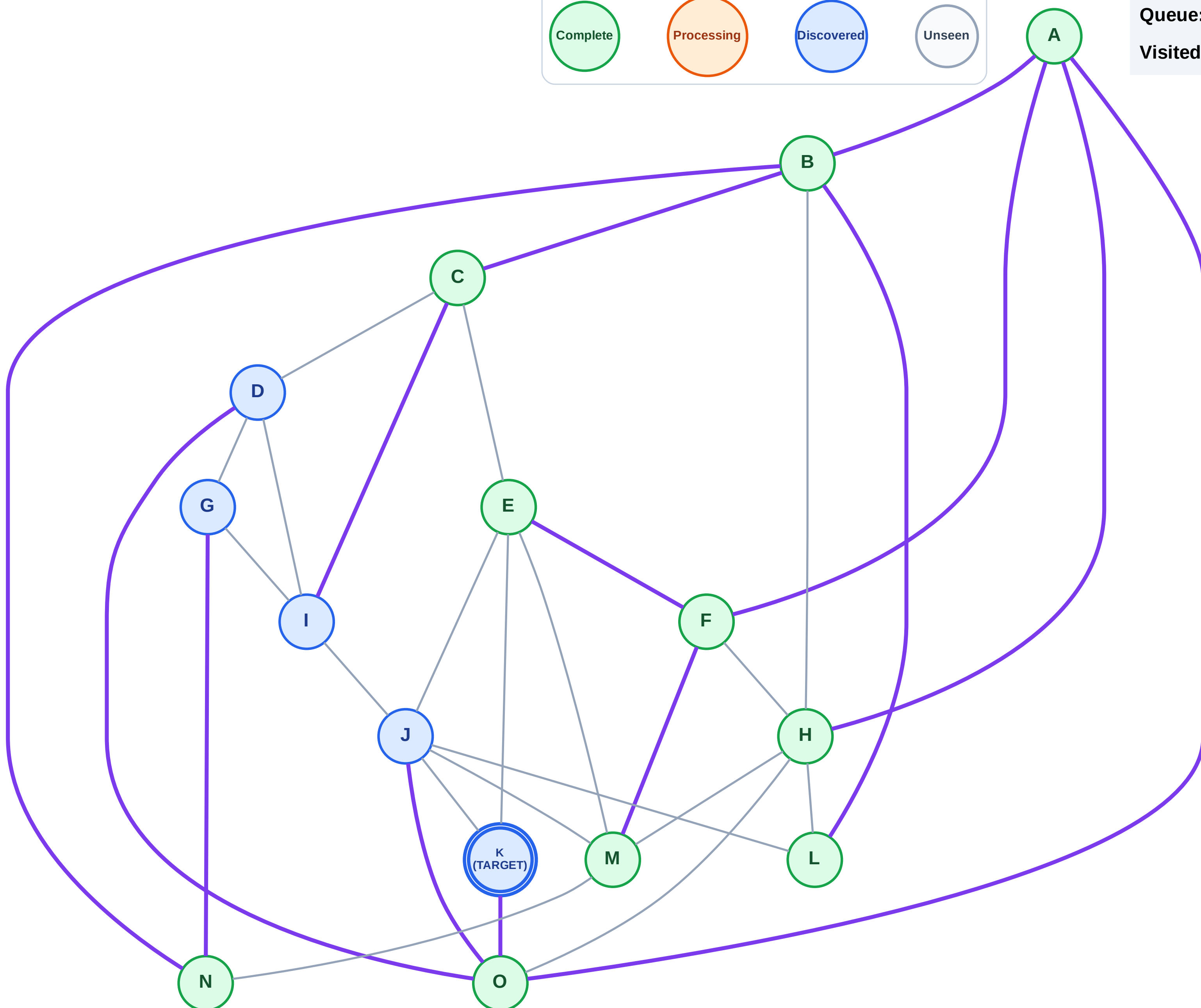


BFS Traversal

Step 21: No new neighbors from M



Queue: D → J → K → I → G
Visited: A, B, C, E, F, H, L, M, N, O



BFS Traversal

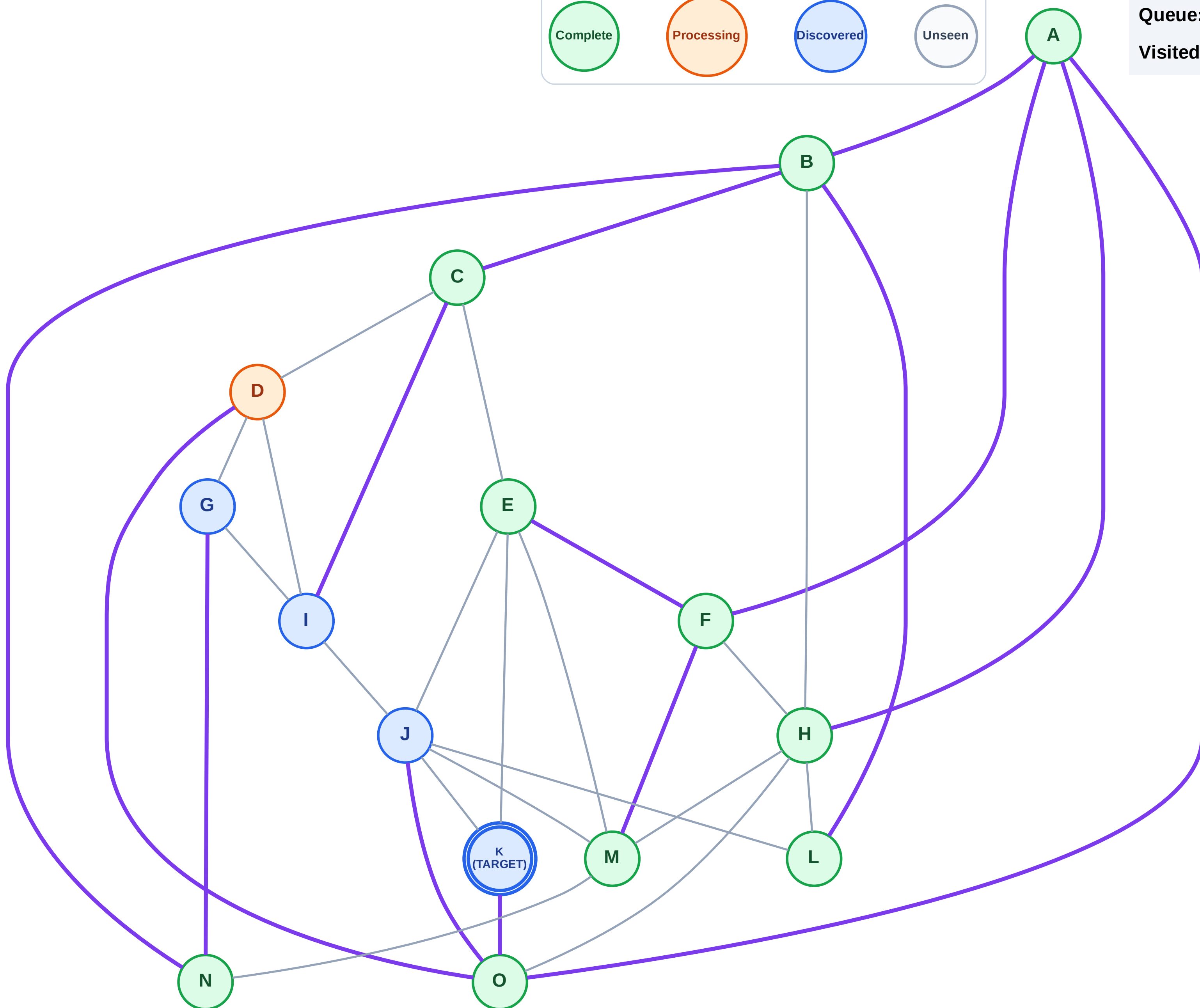
Step 22: Process node D

Legend



Queue: J → K → I → G

Visited: A, B, C, E, F, H, L, M, N, O



BFS Traversal

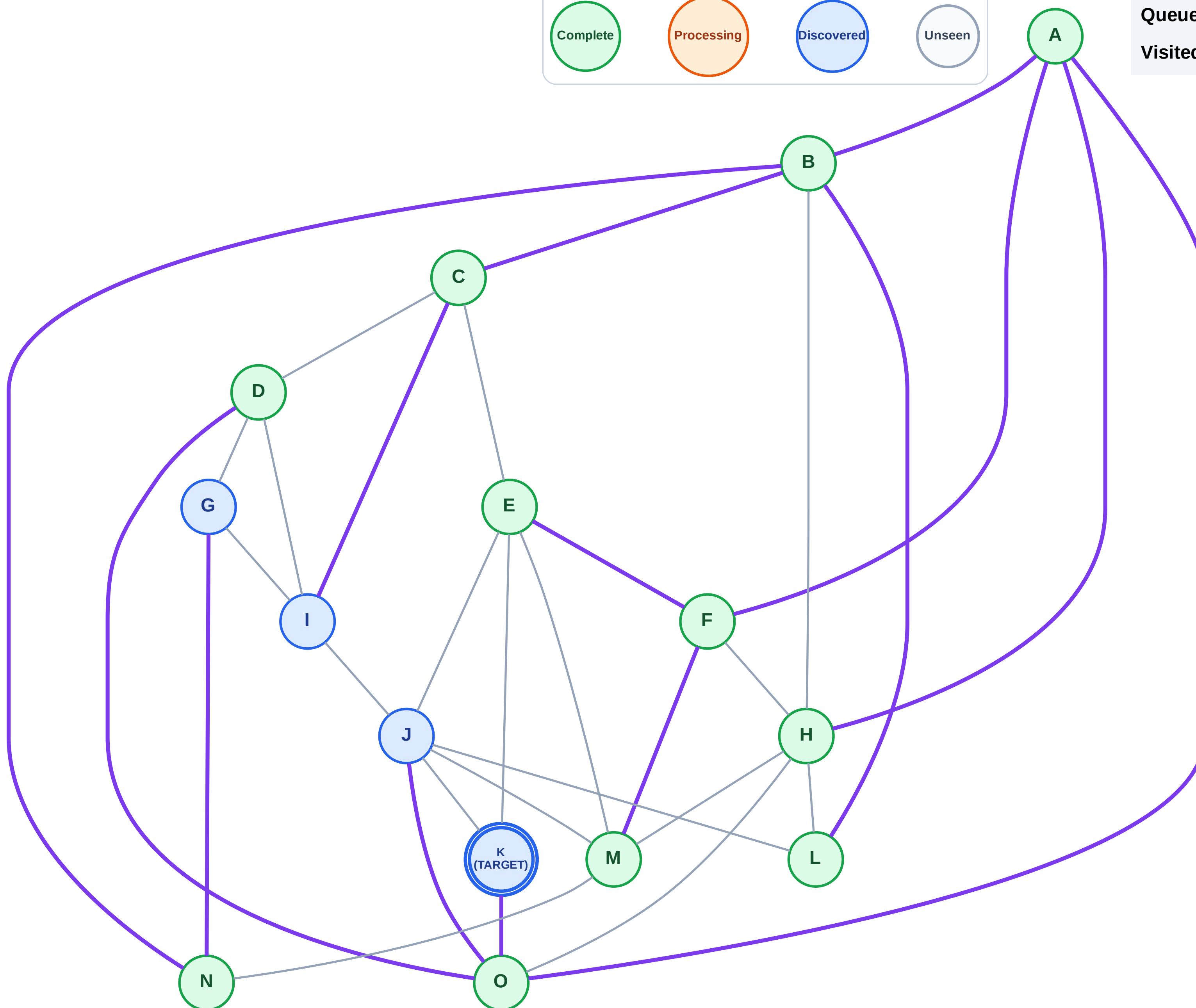
Step 23: No new neighbors from D

Legend



Queue: J → K → I → G

Visited: A, B, C, D, E, F, H, L, M, N, O



BFS Traversal

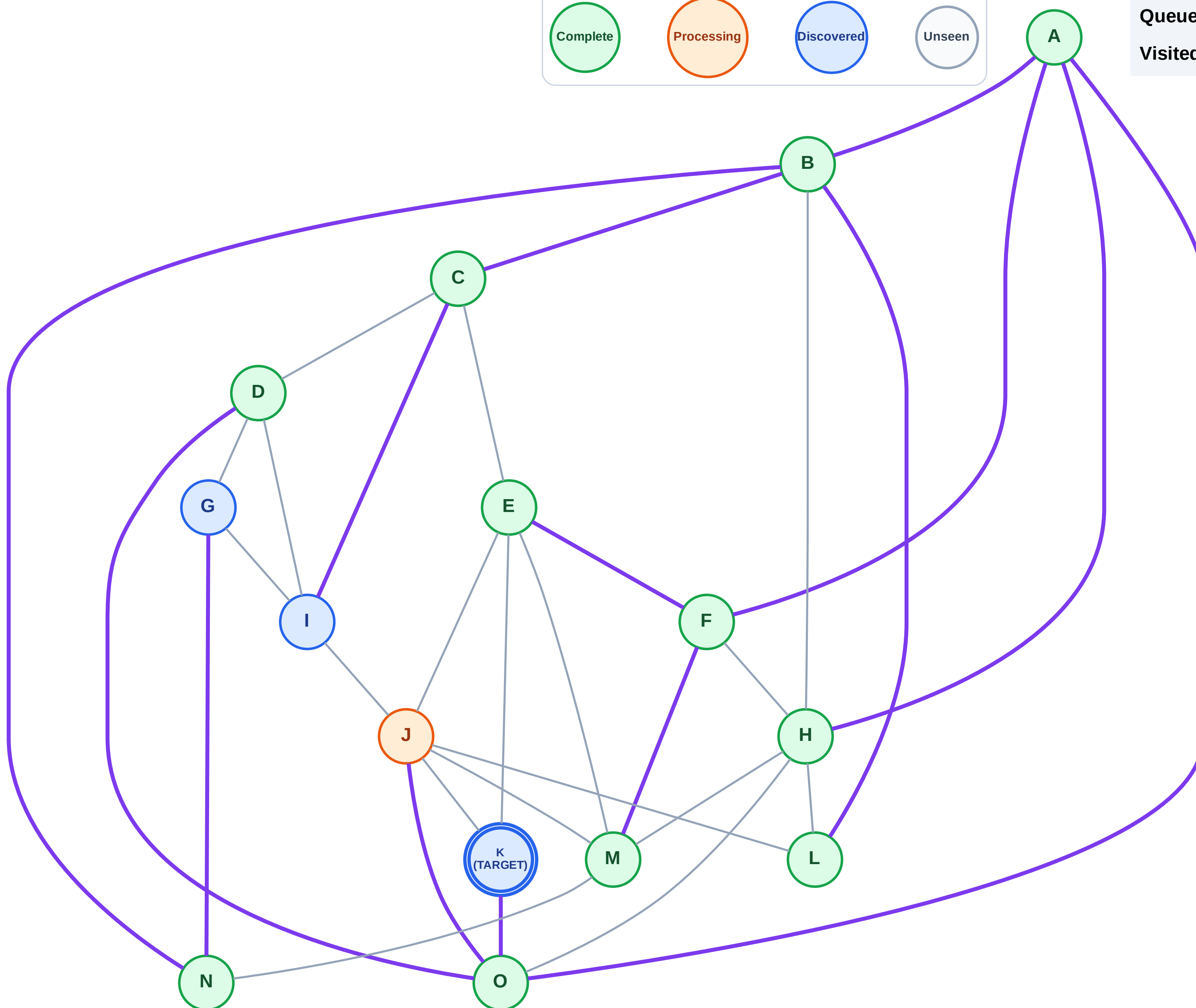
Step 24: Process node J

Legend



Queue: K → I → G

Visited: A, B, C, D, E, F, H, L, M, N, O



BFS Traversal

Step 25: No new neighbors from J

Legend

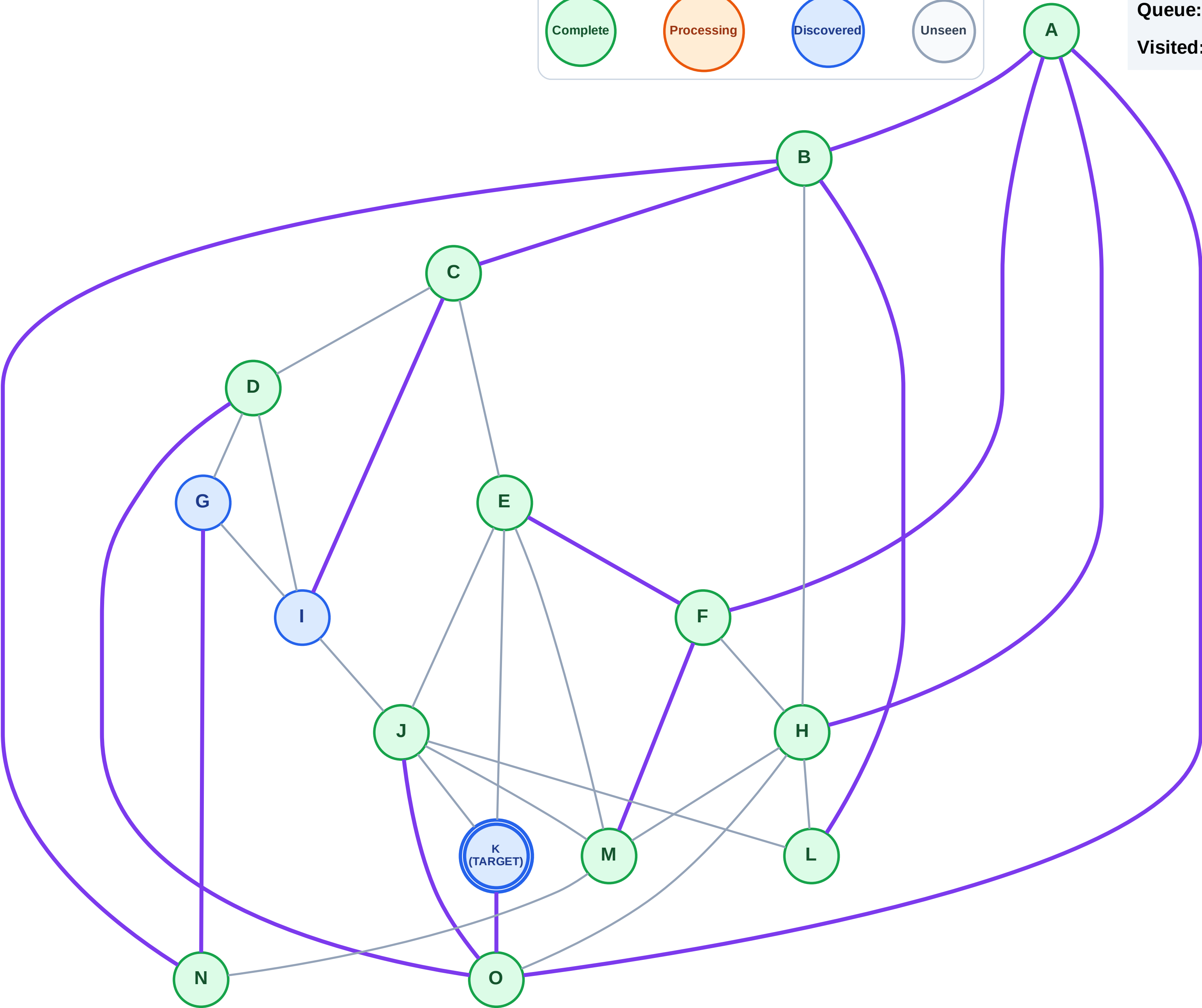
Complete

Processing

Discovered

Unseen

Queue: K → I → G
Visited: A, B, C, D, E, F, H, J, L, M, N, O



BFS Traversal

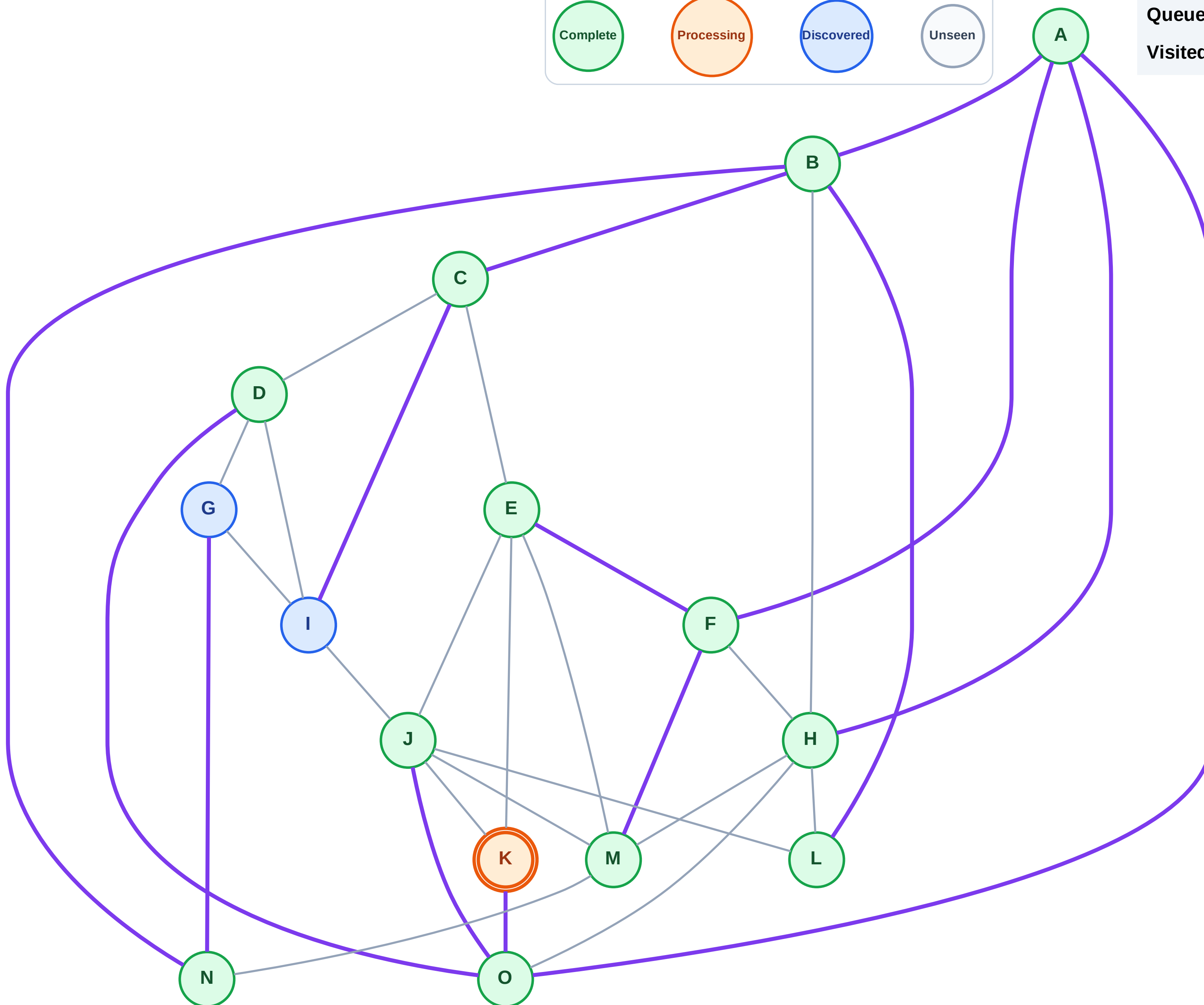
Step 26: Process node K

Legend



Queue: I → G

Visited: A, B, C, D, E, F, H, J, L, M, N, O



BFS Traversal

Step 27: TARGET NODE K REACHED! Path (2 hops): A → O → K

Legend

Complete

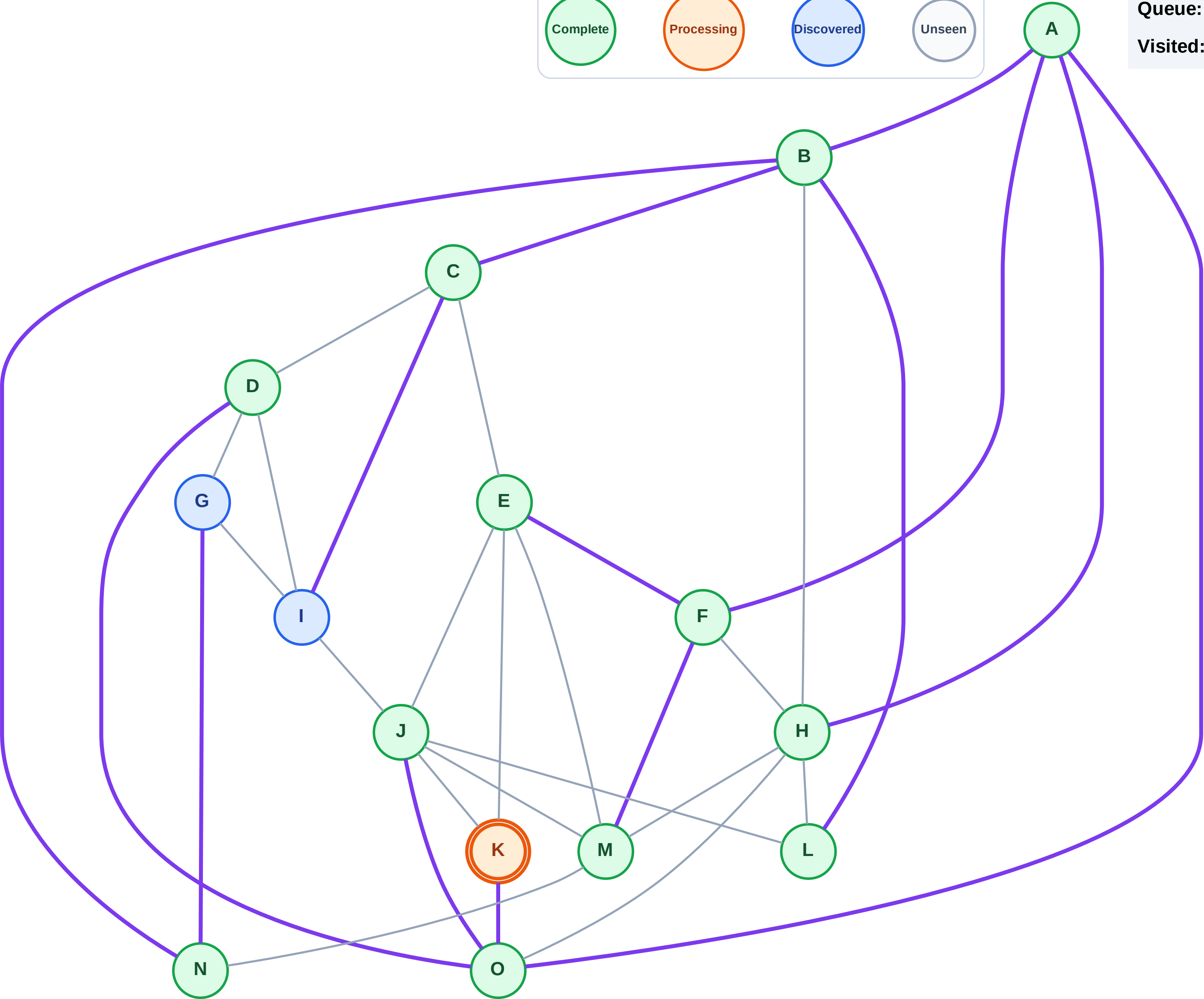
Processing

Discovered

Unseen

Queue: I → G

Visited: A, B, C, D, E, F, H, J, K, L, M, N, O



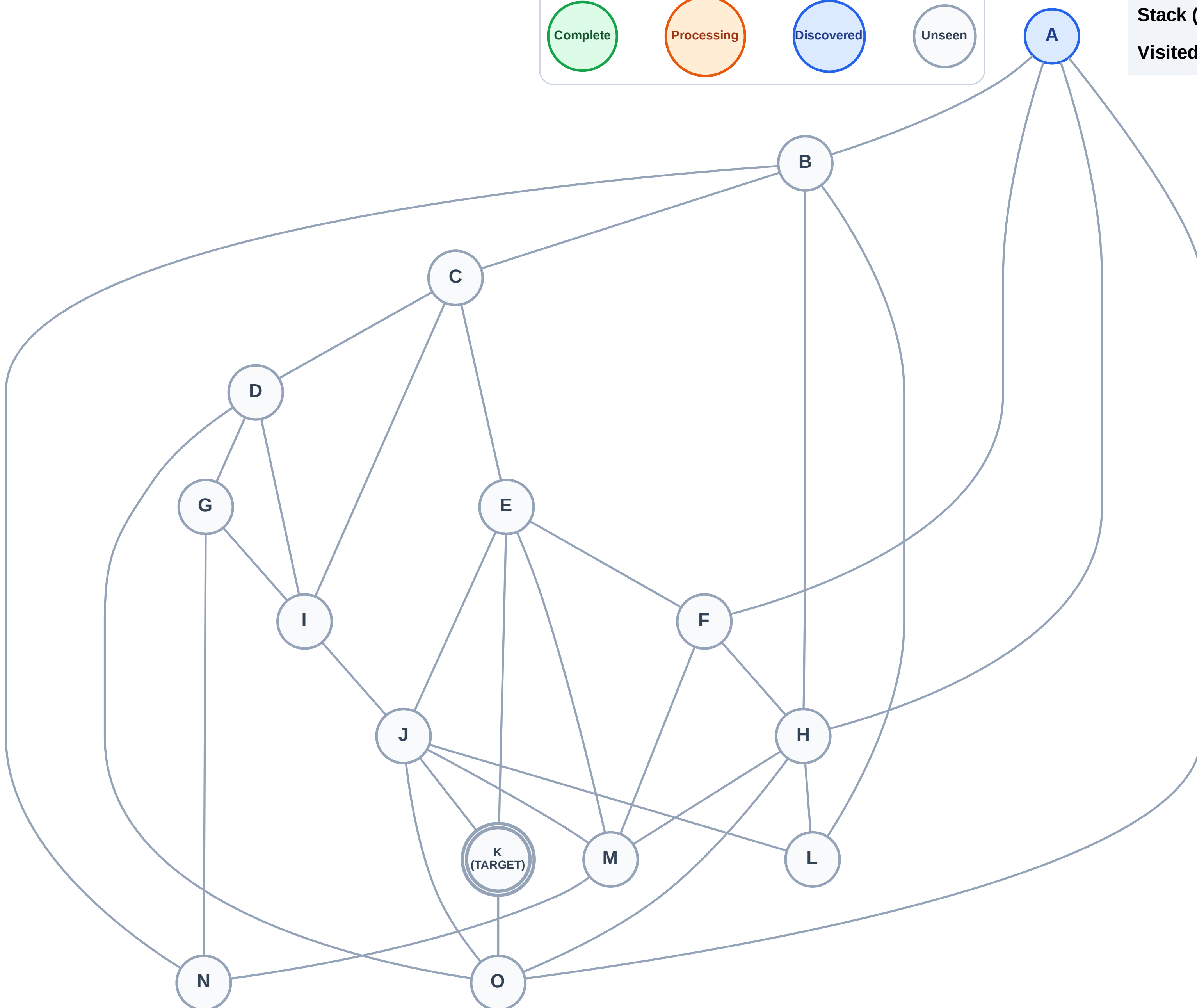
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

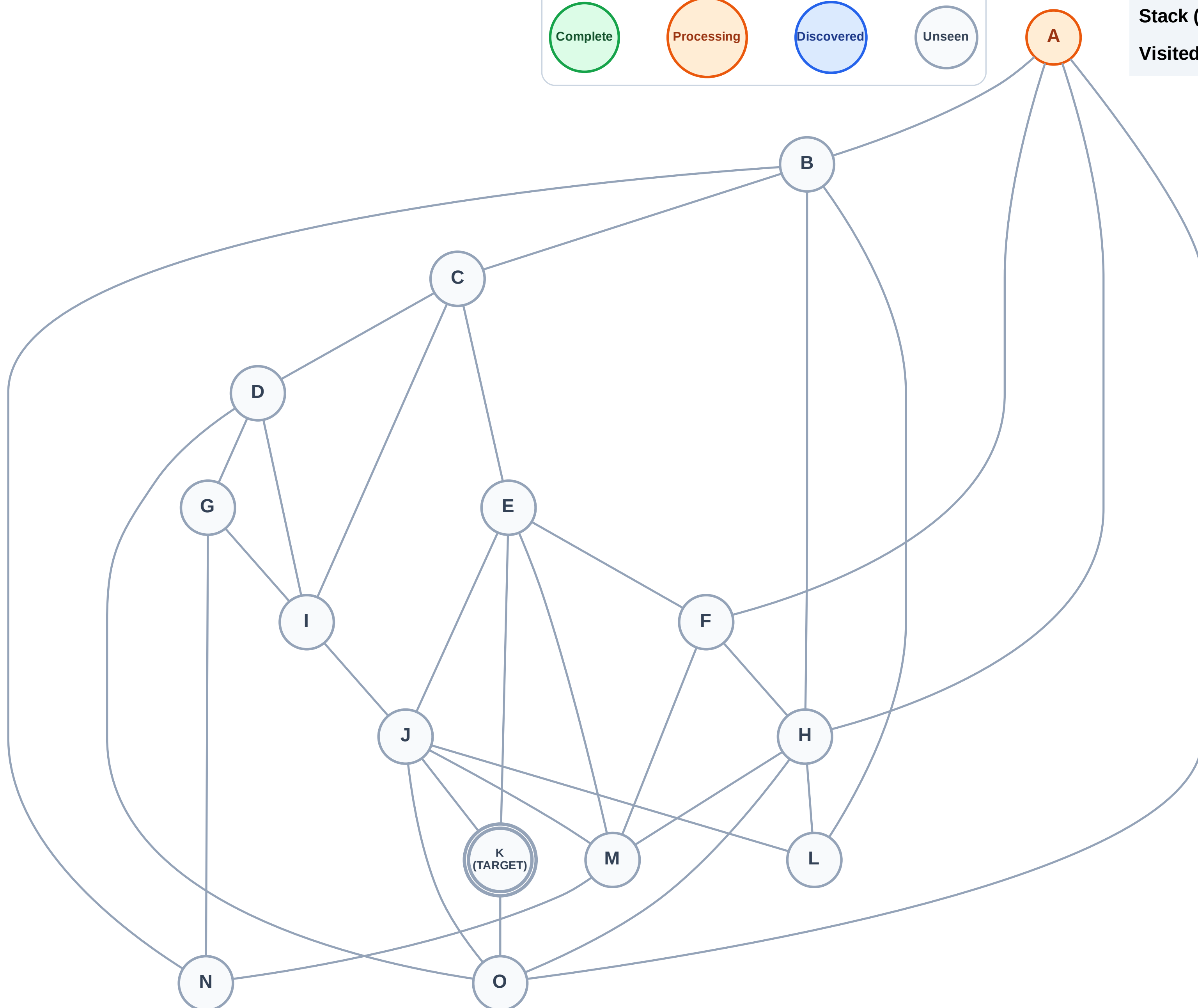
Step 2: Process node A

Legend



Stack (top at right): (empty)

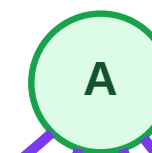
Visited: (none)



DFS Traversal

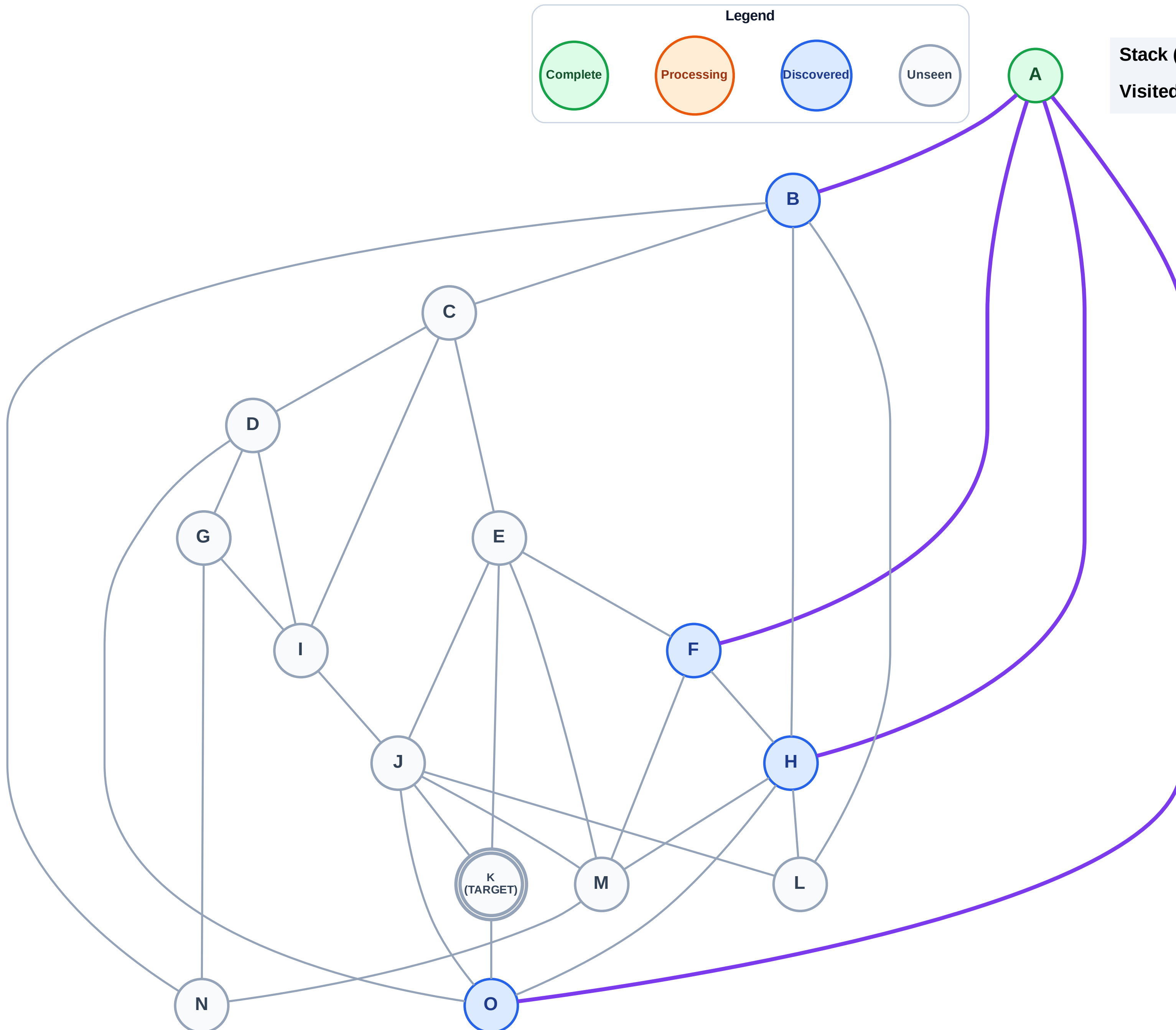
Step 3: Discover O, H, F, B from A

Legend



Stack (top at right): O | H | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

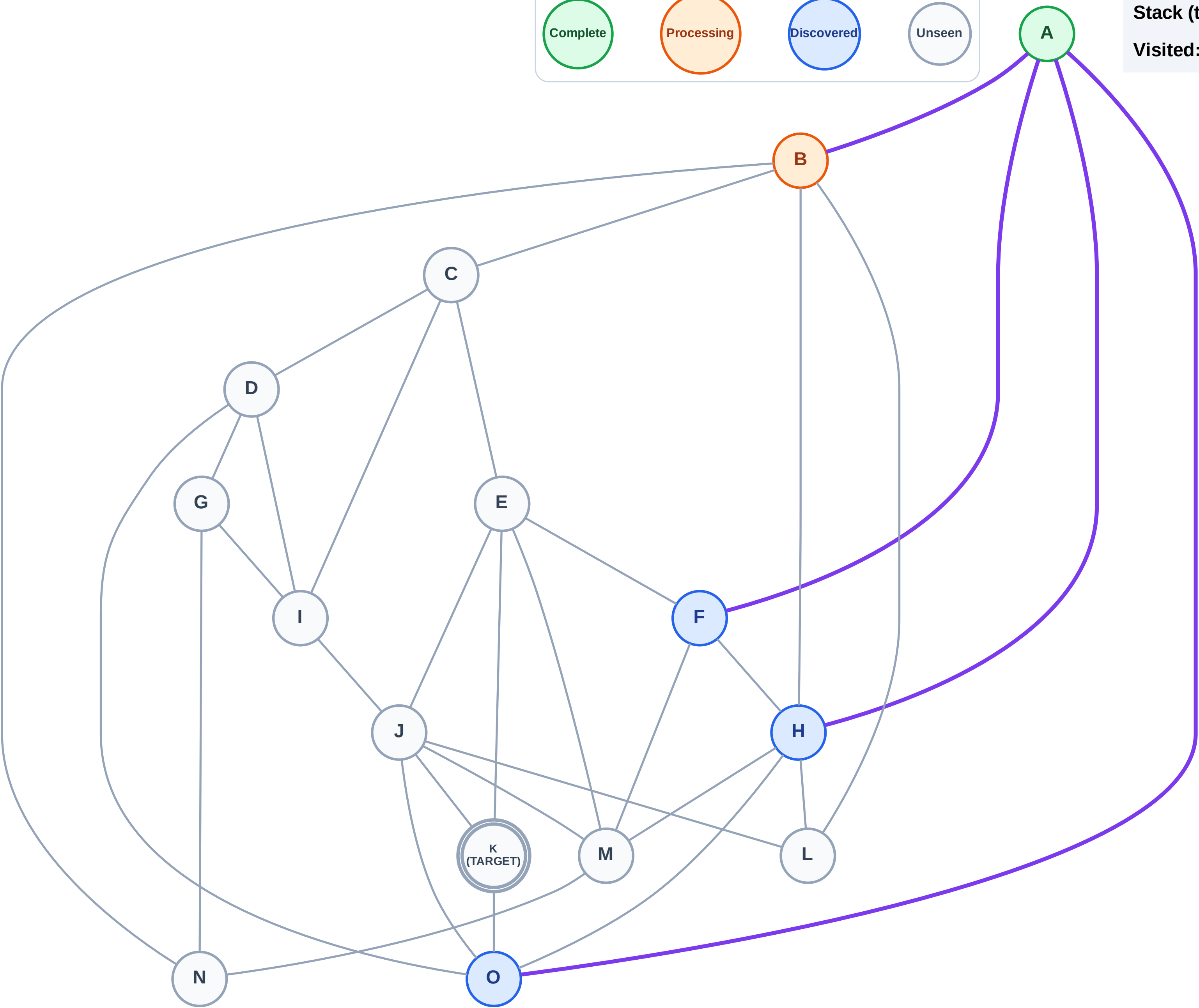
Processing

Discovered

Unseen

Stack (top at right): O | H | F

Visited: A



DFS Traversal

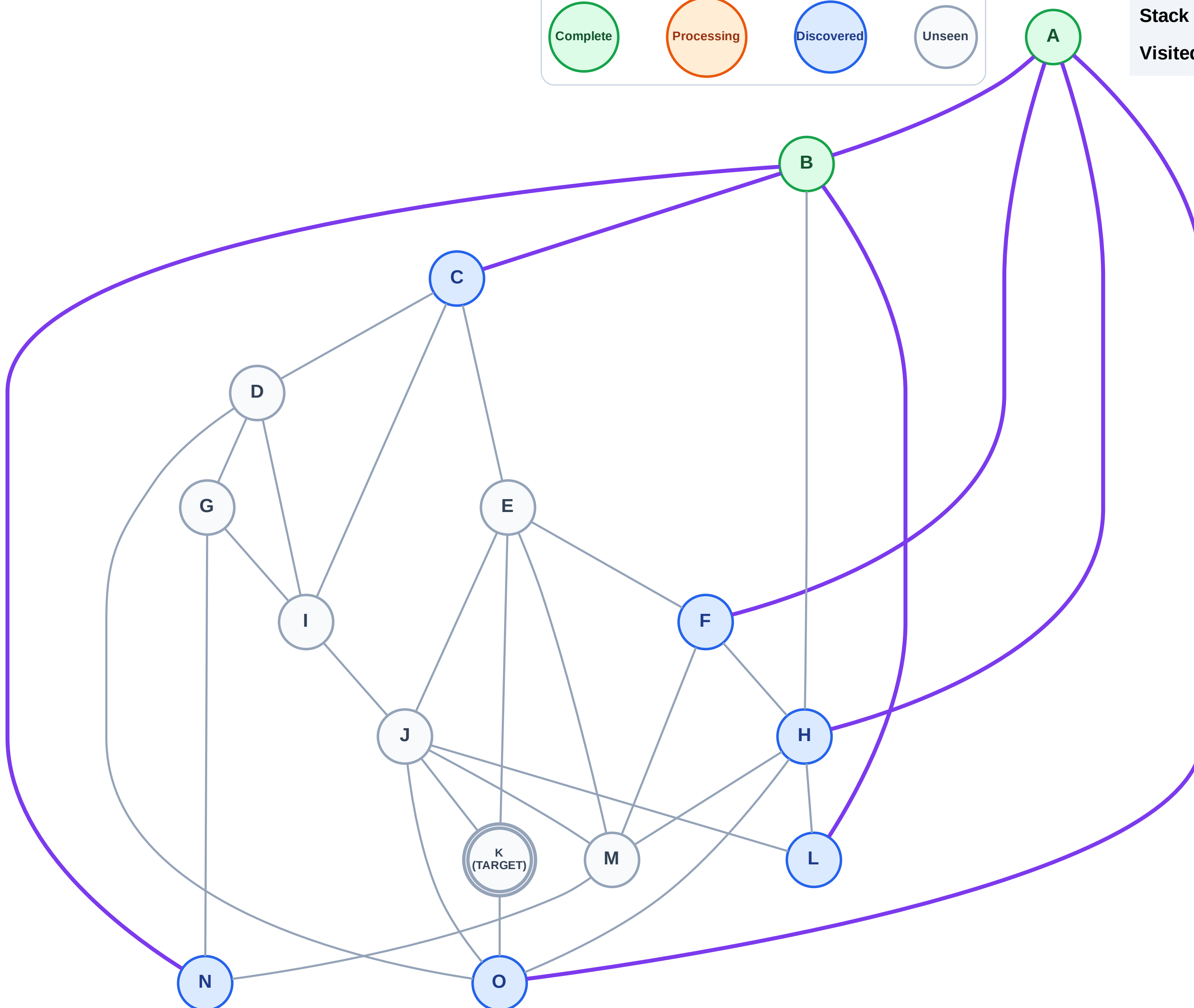
Step 5: Discover N, L, C from B

Legend



Stack (top at right): O | H | F | N | L | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

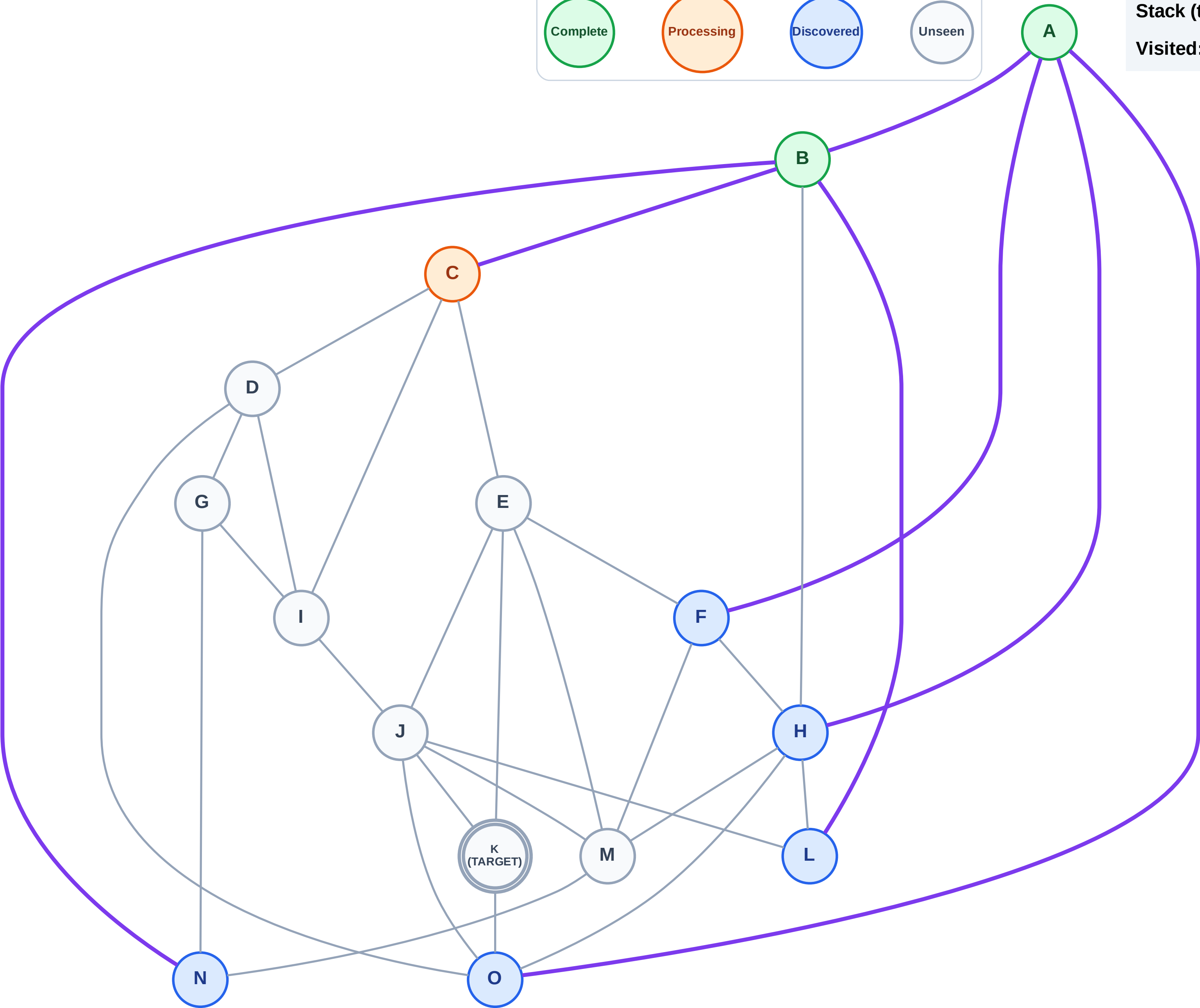
Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L

Visited: A, B



DFS Traversal

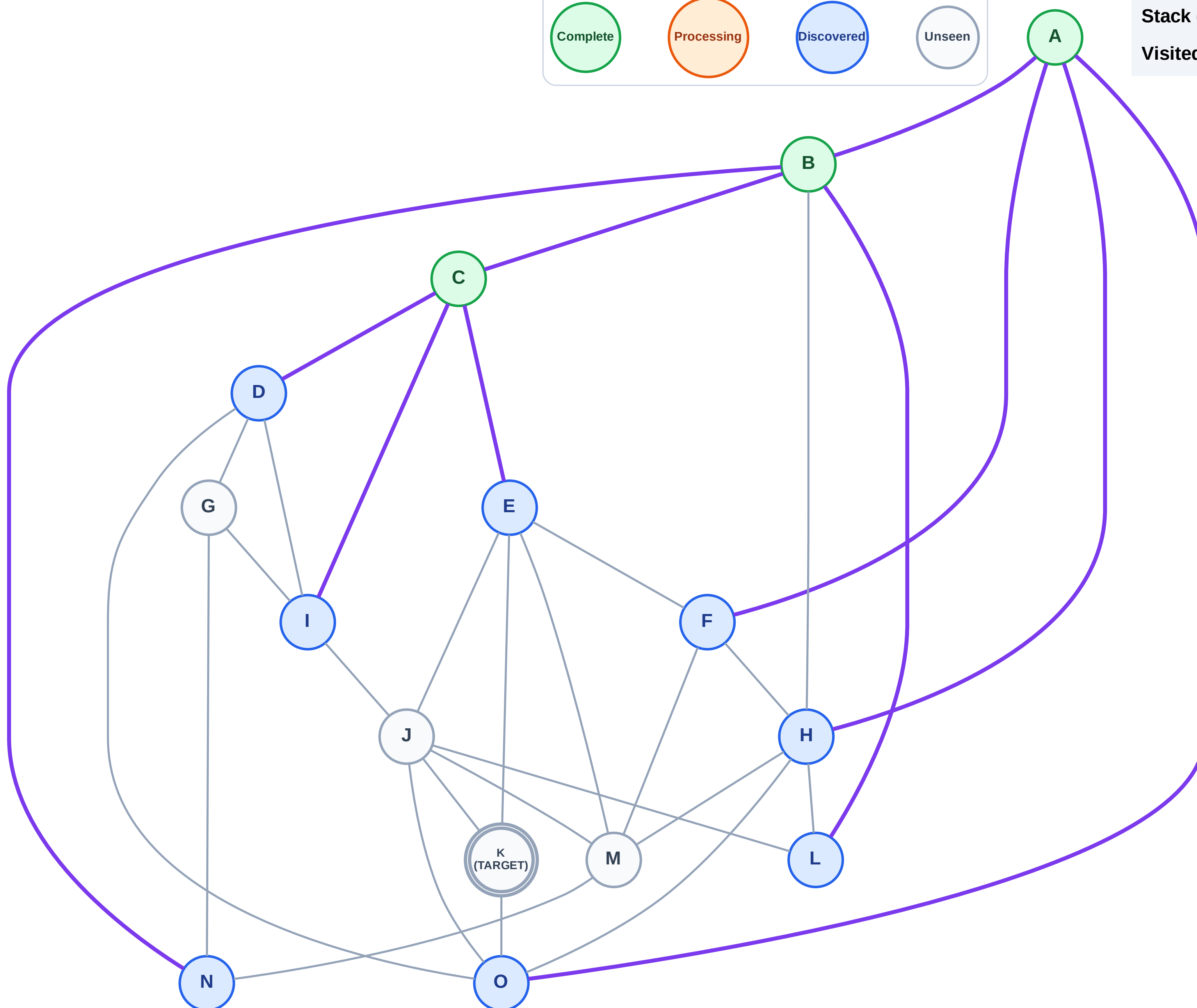
Step 7: Discover I, E, D from C

Legend



Stack (top at right): O | H | F | N | L | I | E | D

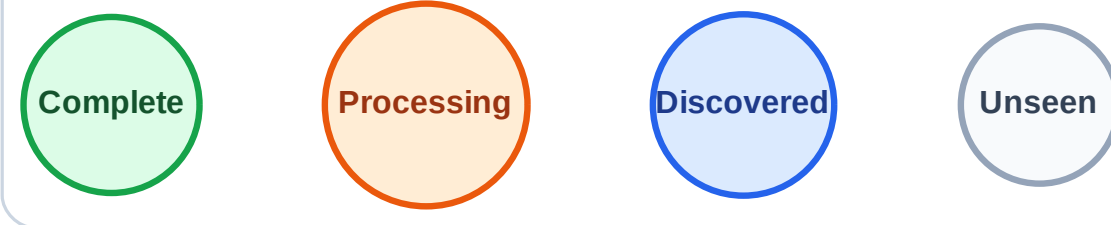
Visited: A, B, C



DFS Traversal

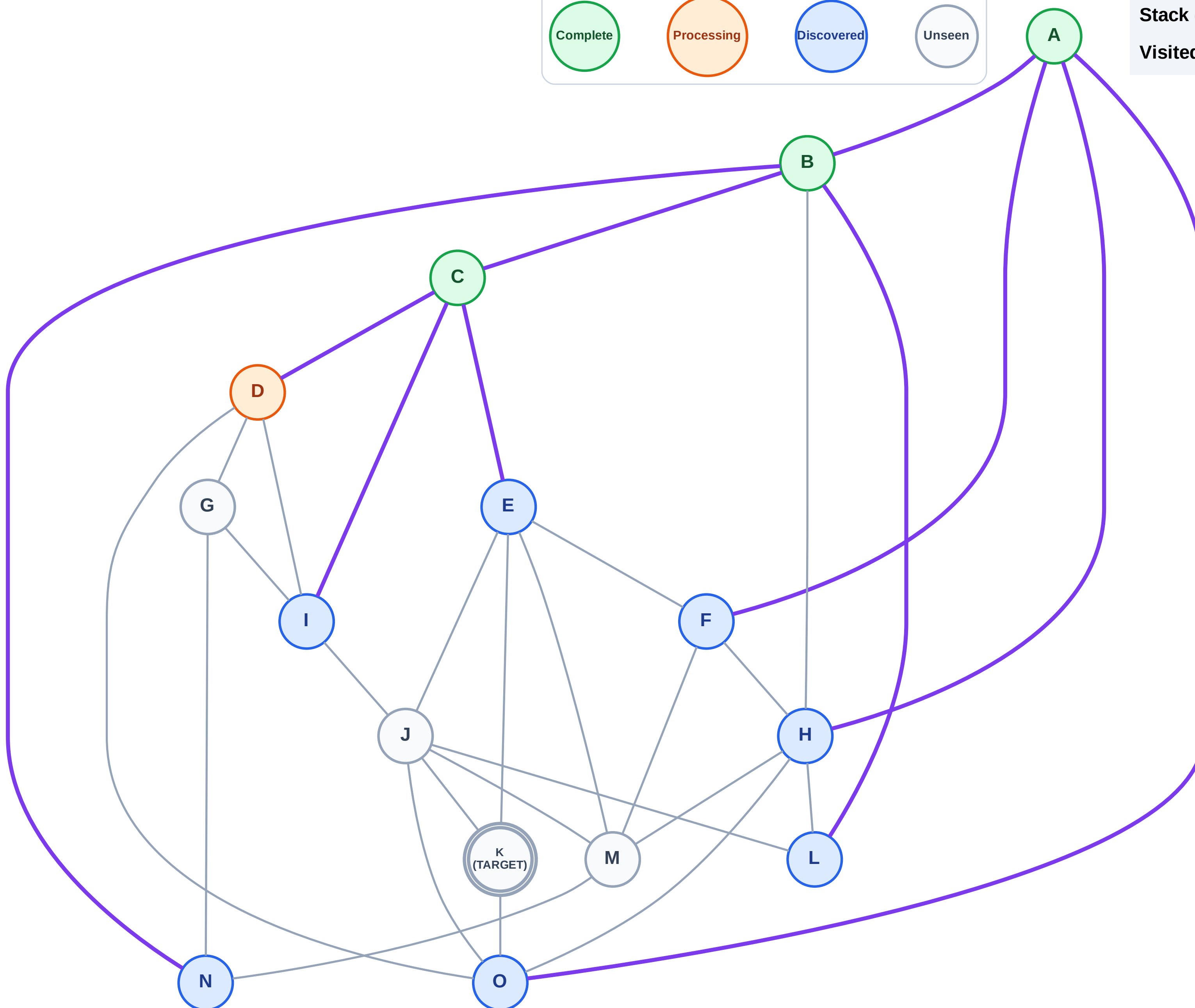
Step 8: Process node D

Legend



Stack (top at right): O | H | F | N | L | I | E

Visited: A, B, C



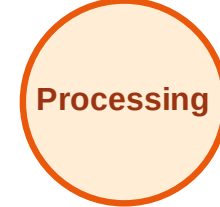
DFS Traversal

Step 9: Discover G from D

Legend



Complete



Processing



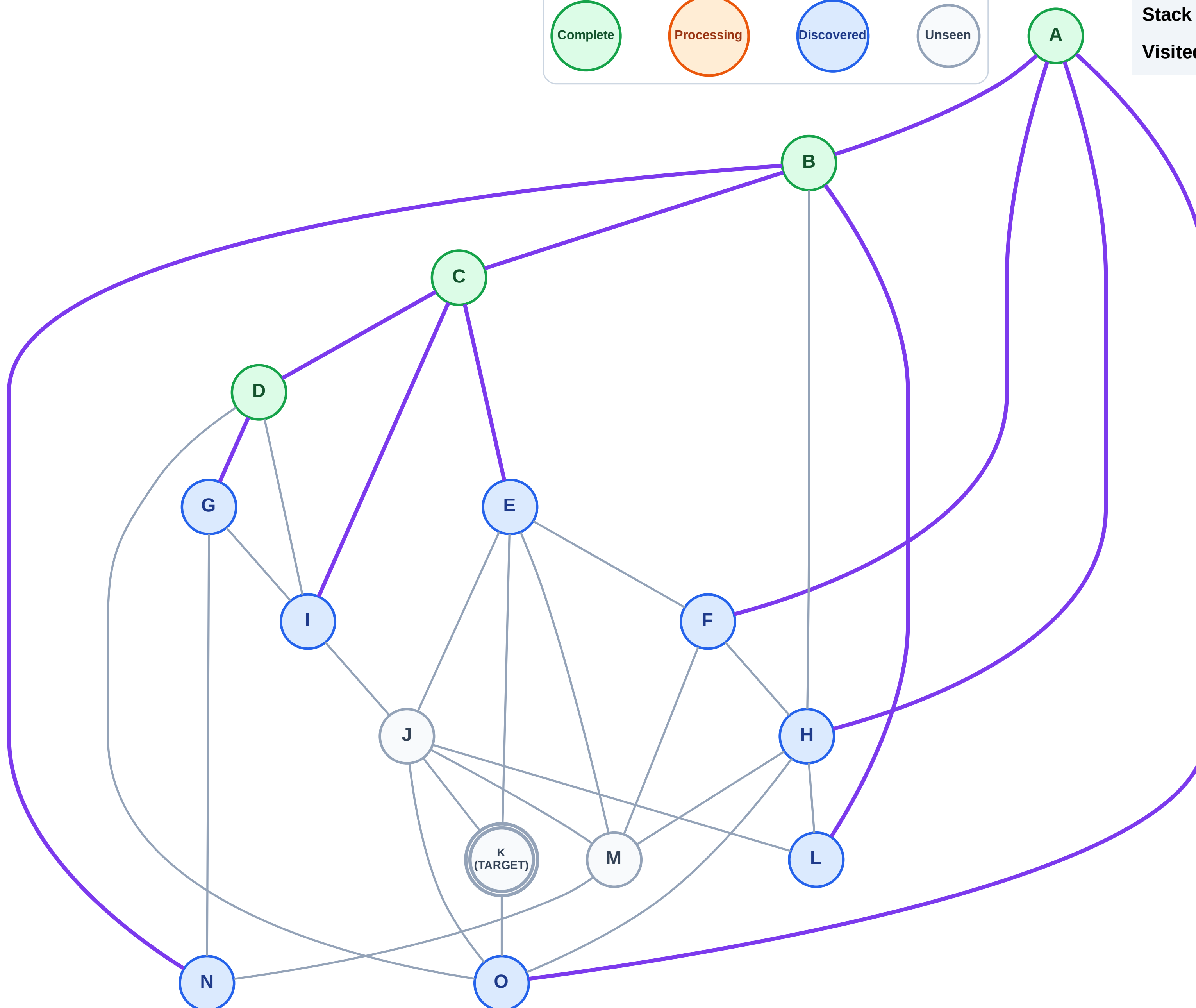
Discovered



Unseen

Stack (top at right): O | H | F | N | L | I | E | G

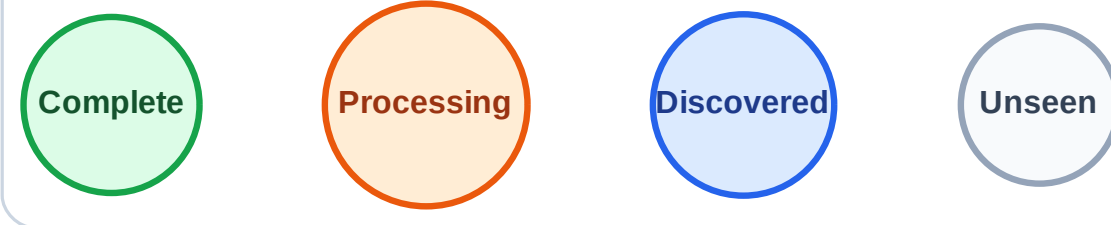
Visited: A, B, C, D



DFS Traversal

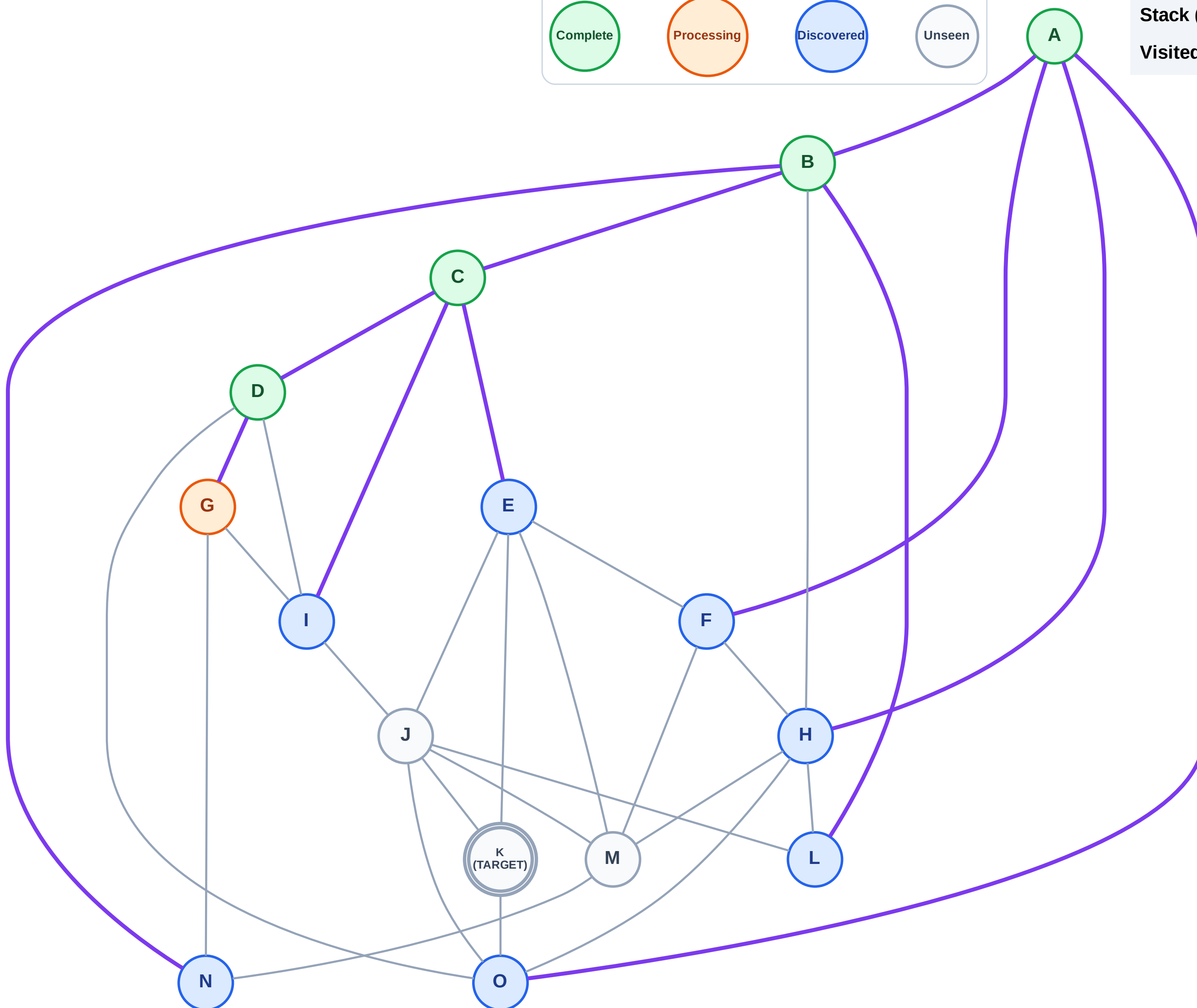
Step 10: Process node G

Legend



Stack (top at right): O | H | F | N | L | I | E

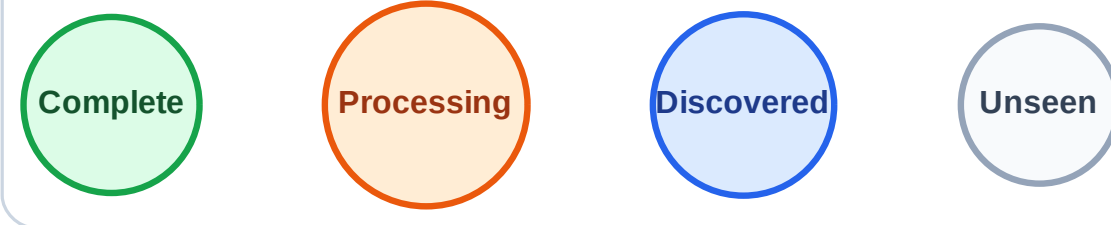
Visited: A, B, C, D



DFS Traversal

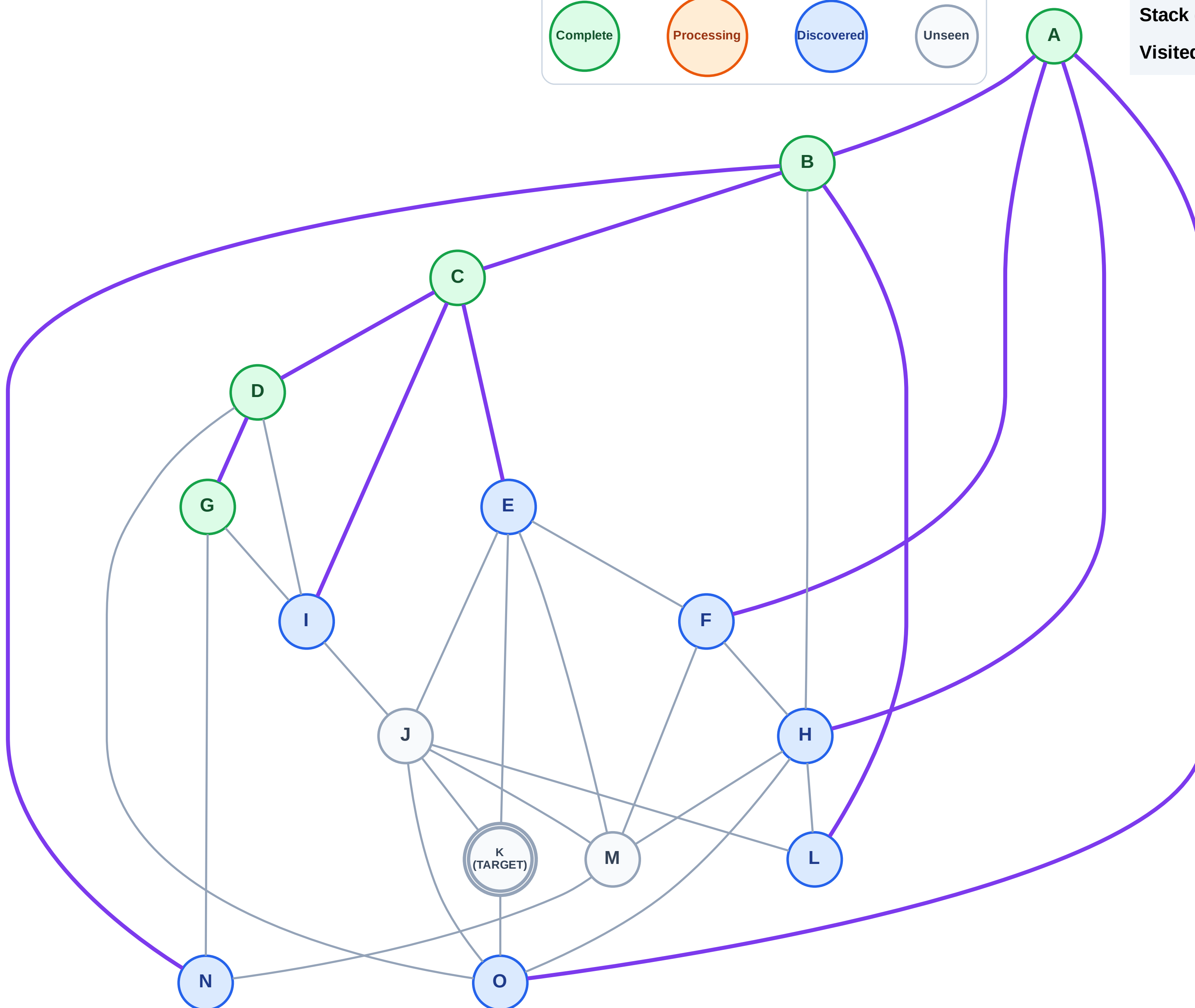
Step 11: No new neighbors from G

Legend



Stack (top at right): O | H | F | N | L | I | E

Visited: A, B, C, D, G



DFS Traversal

Step 12: Process node E

Legend

Complete

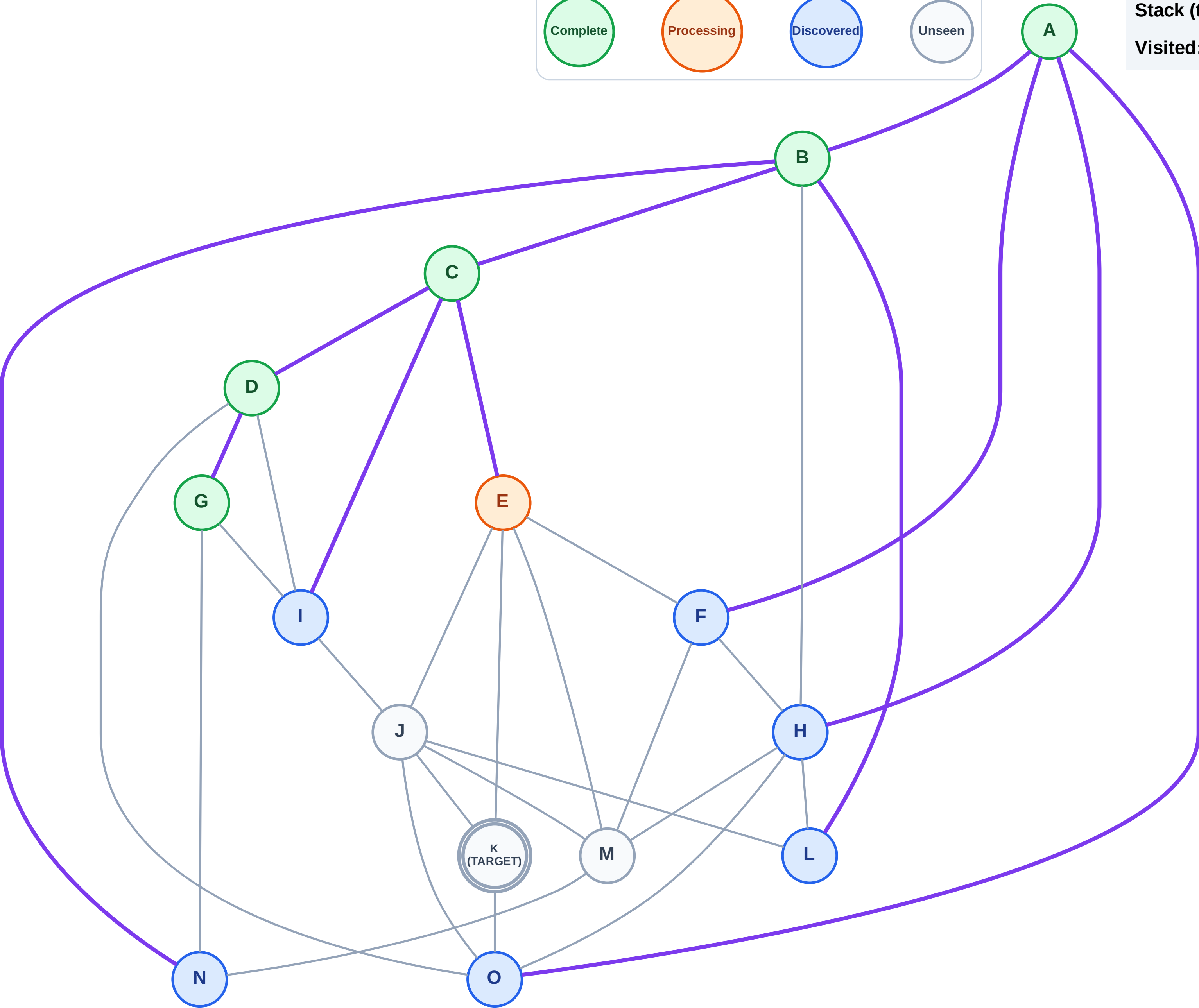
Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L | I

Visited: A, B, C, D, G



Step 13: Discover M, K, J from E

Step 13: Discover M, K, J from E

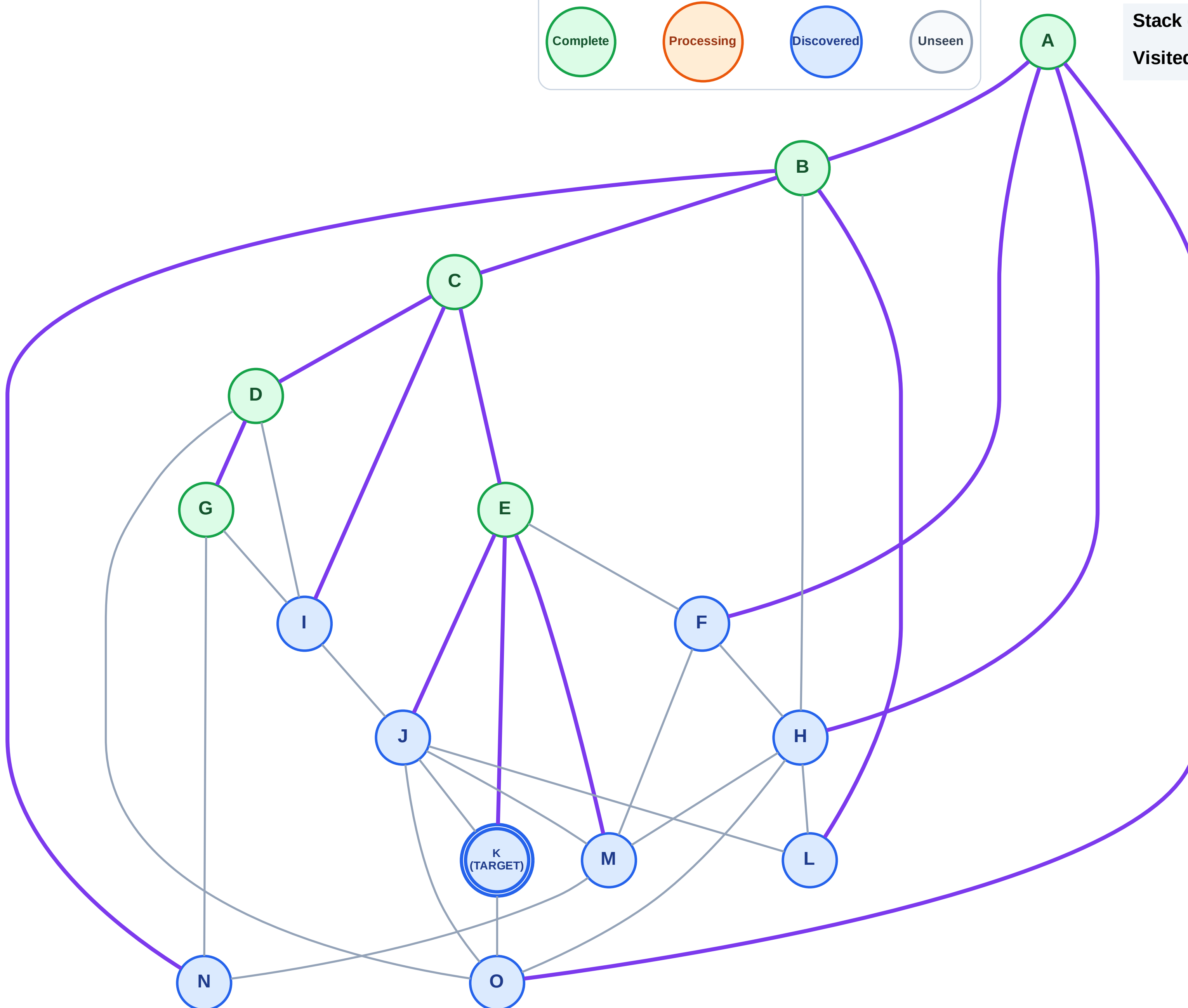
Legend

Complete Processing Discovered Unseen

Processing

Unseen

Visited: A, B, C, D, E, G



DFS Traversal

Step 14: Process node J

Legend

Complete

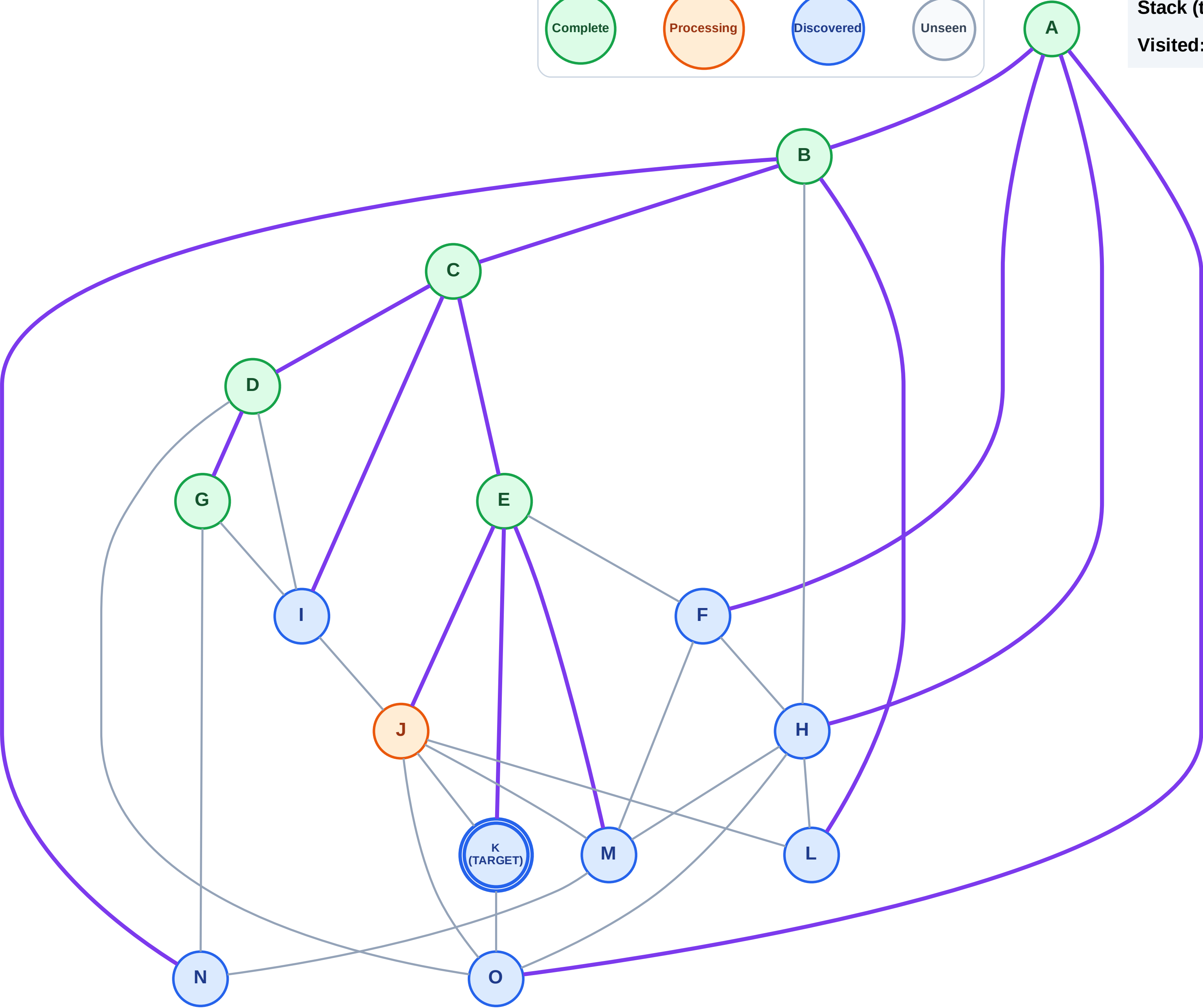
Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L | I | M | K

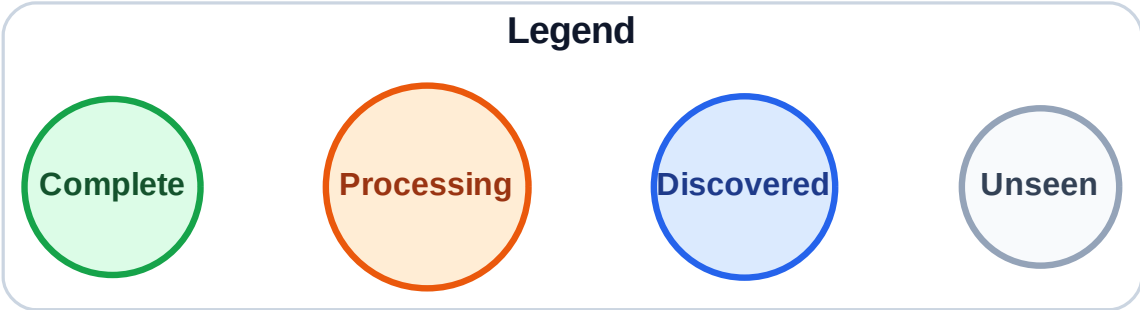
Visited: A, B, C, D, E, G



DFS Traversal

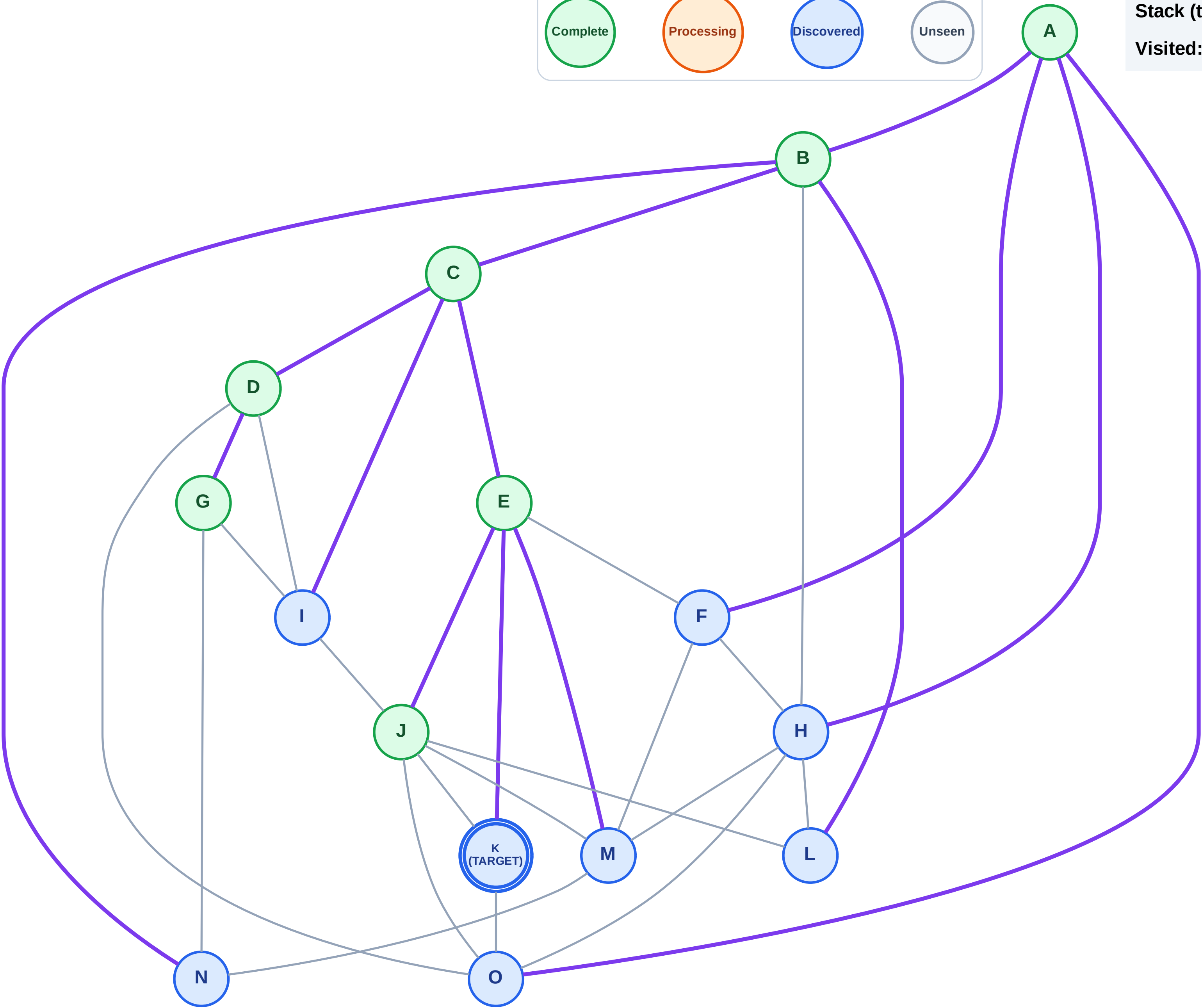
Step 15: No new neighbors from J

Legend



Stack (top at right): O | H | F | N | L | I | M | K

Visited: A, B, C, D, E, G, J



Step 16: Process node K

Step 16: Process node K

Legend

Complete

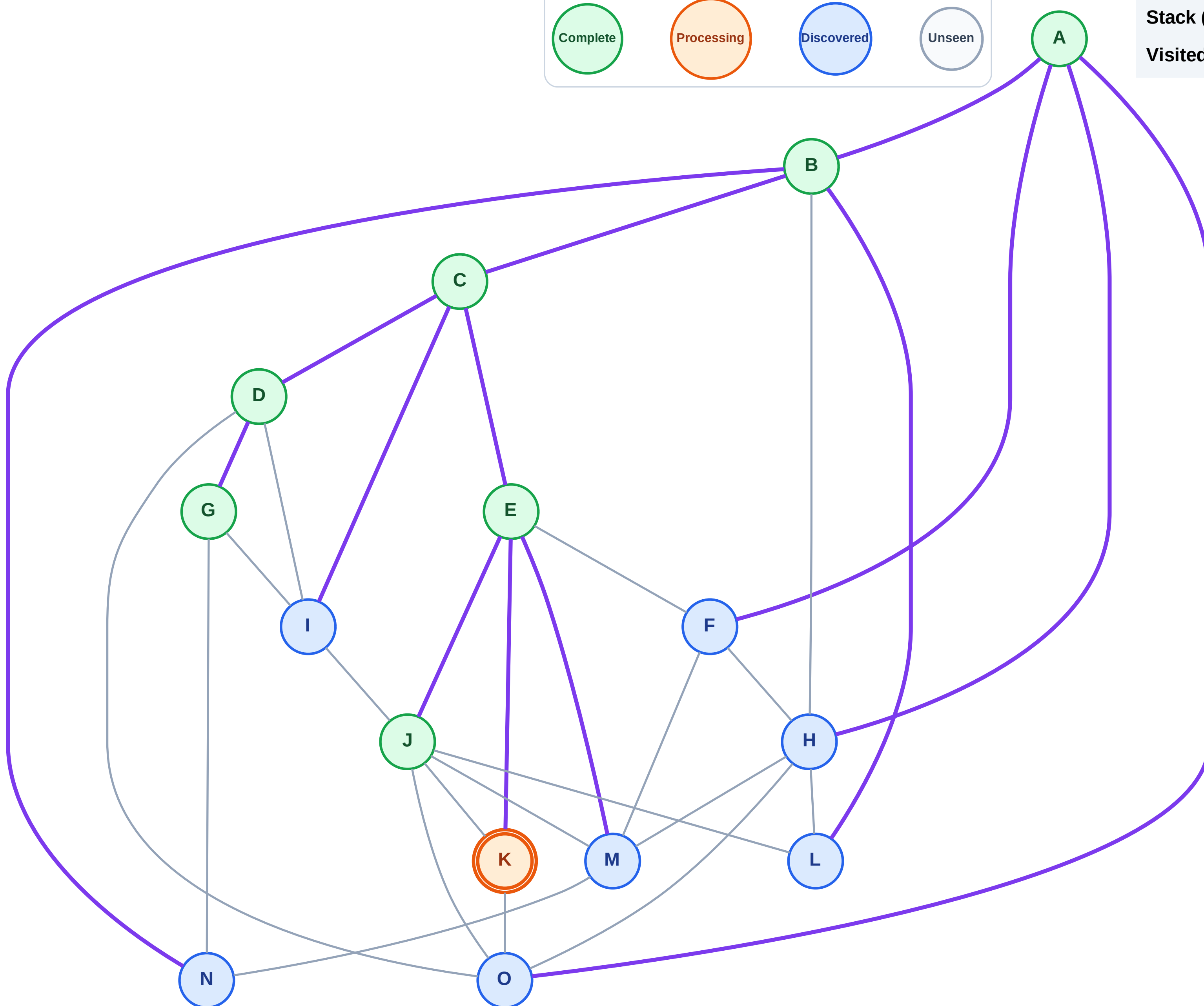
Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L | I | M

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (4 hops): A → B → C → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | H | F | N | L | I | M

Visited: A, B, C, D, E, G, J, K

